Local Gambling Preference and Mortgage Misrepresentation

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Abstract

During the housing boom preceding the Great Recession, higher household leverage was sometimes achieved through mortgage misrepresentation. This paper explores a large sample of mortgages originated between 2005 and 2007, focusing on the misreporting of second liens. We find that second-lien misreporting is more severe in areas with stronger gambling preferences, controlling for other local socioeconomic characteristics and detailed loan and borrower characteristics. In contrast, local factors related to household risk preference, ethics, culture, and financial literacy play limited roles in explaining second-lien misreporting. We also find that second-lien misreporting is more severe when borrowers face higher incentives to misreport. The relationship between second lien misreporting and local gambling preference is robust in subsamples of borrower income, credit scores, and local housing price dynamics. Employing the FICO 620 threshold, we find little evidence that lenders' practices explain the relationship. Finally, we show that the negative performance consequences of second-lien misreporting are amplified in high-gambling-preference areas. We interpret our findings as evidence that borrowers' gambling preferences contribute to the excessive household leverage before the financial crisis.

Keywords: Excessive leverage, Local gambling preference, Second-lien misrepresentation

JEL Codes: G4, G5, R2, R3

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1 Introduction

Household financial decisions have significant effects on the aggregate market (Campbell, 2006). A prominent example is the rapid expansion of U.S. household leverage, which played a central role in the 2008 financial crisis. From the expansion of credit supply to overly optimistic expectations about house prices, researchers find that households across all income groups took on excessive leverage in the mortgage market (Mian and Sufi, 2009; Adelino et al., 2016)[1]. While much of this literature debates the underlying drivers, distorted beliefs versus misaligned incentives, relatively little attention has been paid to heterogeneity across households in such an aggressive borrowing environment. This paper provides empirical insights into which borrowers were more likely to take on excessive leverage.

In the years preceding the crisis, a striking example of excessive leverage was the misrepresentation of second liens, in which the existence of a simultaneous second lien was misreported at the time of first-lien origination (Piskorski et al., 2015; Griffin and Maturana, 2016). A standard first lien mortgage already places a household in a leveraged position, and the addition of a second lien further increases leverage by reducing homeowner equity. Yet loans with such high leverage were often difficult to sustain, especially when they combined high second lien loan to value ratios, risky first mortgages, or low documentation features. Yavas and Zhu (2024) show that the concealment of second liens was concentrated in these fragile loans, which were perceived as harder to sell in secondary markets. This finding suggests that, under truthful reporting, such loans would have been less attractive for lenders to approve because of their heightened risk and limited marketability. By hiding the second lien, borrowers were able to make their applications appear less risky and more likely to be approved. In this way, misreporting allowed households to obtain leverage that would have been difficult to achieve under truthful reporting. Therefore, we focus on the second lien misrepresentation in this study.

[1]Mian and Sufi (2009) show that subprime borrowers gained excessive access to mortgage credit that became disconnected from income growth, while Adelino et al. (2016) argue that middle- and high-income borrowers also increased leverage substantially and contributed to the rise in delinquencies during the crisis.

We start by investigating how local socioeconomic characteristics, including social environment, economic status, local market conditions, and household structure, shape the incidence of second-lien misrepresentation. After controlling for other local characteristics as well as detailed loan and borrower characteristics, we find that second-lien misrepresentation is more prevalent in areas with higher Catholic-to-Protestant ratios (CPRATIO), which we interpret as a proxy for local gambling preferences following Kumar et al. (2011). In contrast, other local factors associated with household risk preferences, ethics, culture, and financial literacy, such as religiosity and education, play limited roles in explaining second-lien misreporting.

Our interpretation is that local gambling preference affects borrowers' financial decisions, making them more willing to engage in opportunities with right-skewed payoffs. From households' perspective, second-lien misrepresentation resembles such an opportunity. The upside is obtaining better loan terms or gaining access to homeownership, which is emotionally tied to the American Dream and financially offers the possibility of wealth accumulation[2]. If house prices rise strongly, the gains can be very large, while if prices rise only moderately or stagnate, the returns are smaller and may even turn negative. At the time, however, most households and lenders believed that house prices would continue to rise, so the expectation of flat or negative returns was uncommon. The downside is limited to the costs of application, the effort of switching lenders, and the relatively weak legal penalties before the crisis[3]. This combination of potentially very large gains with relatively modest losses produces a right-skewed payoff structure that is attractive to gambling-prone borrowers.

The borrower's gambling preference interpretation implies that the effect should be stronger when the motives to misreport are stronger. Christensen et al. (2018) show that local gambling attitudes increase corporate misreporting, and that the effect is amplified when

[2]See Clinton (1995); Bush (2003) for how homeownership is part of the American Dream. See Bostic and Lee (2008); Goodman and Mayer (2018) for how, with appropriate mortgage terms, homeownership brings large financial returns and helps households accumulate wealth.

[3]Loan denial was common, but the cost of loan application and the cost of switching lenders were relatively low. Legal penalties (Henning, 2009; Federal Housing Finance Agency, 2023) were usually limited pre-crisis when law enforcement focused on fraud for profit rather than fraud for housing (FBI, 2006, 2007).

the motives to misreport are stronger. Following this logic, we examine subsamples that vary borrowers' motives to obtain the loan using property occupancy and loan purpose. Motives are stronger for primary residences because loan approval provides essential housing, while motives are weaker for non-primary residences that are often discretionary or used for investment. Motives are also stronger for purchase loans because approval is required to acquire the property, whereas refinancing borrowers already own the home. Consistent with this reasoning, we find that the positive effect of local gambling preference on misrepresentation is more pronounced in these high motive subsamples.

We also examine subsamples defined by borrower income, credit scores, and local housing price dynamics. These dimensions are connected to debates about the origins of the financial crisis (Adelino et al., 2018), but they also capture variation in borrowers' incentives to misreport. We find that the effect of local gambling preference is present across subsamples and is stronger where borrower incentives are higher. These results are consistent with the interpretation that gambling preference shapes misrepresentation behavior and complement existing explanations of the crisis.

One concern with the interpretation is that local gambling preference may capture not only borrower behavior but also lender practices, since lenders were likely aware of the existence of second-lien misrepresentation (Piskorski et al., 2015). To address this, we use the Keys et al. (2010) setting, where the ease of securitization for low-documentation loans with FICO scores at or above 620 provides a shock to lenders in screening and potentially facilitating misrepresentation(Griffin and Maturana, 2016). We implement a regression discontinuity design (RDD) to estimate jumps in the number of loans and default rates, and apply a difference-in-discontinuities (diff-in-disc) method to compare these across high and low gambling preference areas. In the pooled sample, lenders appear to facilitate simultaneous seconds in general, but not specifically misreported seconds, as the jump ratios are smaller for misreported than for correctly reported seconds. Moreover, the default-rate discontinuity is insignificant for misreported seconds, suggesting that lax screening was not more pronounced for such loans. Importantly, when comparing high and low gambling preference areas, jumps in both the number of loans and default rates for misreported seconds are smaller in high gambling preference areas, implying that lenders in these areas do not play a larger role in facilitating misrepresentation. Taken together, these results provide little evidence that lender practices explain the relationship between gambling preference and misrepresentation.

Finally, given this gambling-based interpretation, we examine whether the performance of misreported loans differs across gambling environments. Misreported second liens are generally associated with poorer loan performance, and we find that this relationship is substantially stronger in high gambling preference areas. This pattern indicates that local gambling preference amplify the fragility of excessively leveraged loans. A back-of-the-envelope calculation shows that the heightened default risk of misreported loans in gambling-prone areas corresponds to roughly \$1.25 billion in additional first-lien exposure, underscoring the economic importance of gambling preference in the mortgage market.

To reinforce the findings and interpretation, we conduct a series of robustness checks. First, to address functional form concerns, we re-estimate the models using different specifications and find consistent results. The probit models account for the binary nature of the dependent variable, while the causal forest approach of Wager and Athey (2018) further strengthens causal inference. Second, to reinforce the gambling preference interpretation, we examine alternative measures of gambling propensity and find similar results. Third, to validate the discussion of the lender channel, we confirm that the RDD and difference in discontinuities are valid and that the results are not sensitive to specification choices. Fourth, to address the concern that the use of jumps in number of loans is an aggregate measure that does not incorporate the loan-level controls of the baseline regressions, which may create inconsistency across specifications rather than rejecting the lender channel, we instead use the misrepresentation rate as the outcome and re-estimate the models with the full set of controls and fixed effects. The results remain consistent, further supporting our interpretation. Finally, to address outcome measurement, we examine loan performance using alternative definitions of default and apply causal forest analysis, both of which confirm that the results persist.

Our paper contributes to several strands of literature. First, it emphasizes the importance of heterogeneity in borrower characteristics, especially behavioral traits, in understanding household decisions to take on excessive leverage in the years leading up to the financial crisis. Prior research highlights how credit supply expansion and distorted house price beliefs contributed to households taking on excessive leverage (e.g., Mian and Sufi, 2009; Adelino et al., 2018; Justiniano et al., 2019). While these studies focus on aggregate forces that broadly shaped household borrowing, my paper highlights behavioral heterogeneity by showing that differences in local gambling preferences help explain why some households were more willing to take on excessive leverage than others.

Second, our paper complements the literature by discussing where in the credit supply chain (from borrowers to MBS underwriters) second-lien misrepresentations took place. By studying privately securitized loans, Griffin and Maturana (2016) point out that lenders were likely aware of the misrepresentation, attributing such misreporting more to lenders than to underwriters. Conversely, Piskorski et al. (2015) show that underwriters could easily uncover the misrepresentation and also play an important role in this issue. Using both portfolio loans and securitized loans, Yavas and Zhu (2024) support the argument that second-lien misrepresentation occurs in the early stages of intermediation by lenders, while underwriters have limited but significant effects on reducing the occurrence of misrepresentation through screening. Our paper examines this issue from the borrower's perspective, showing that borrowers also significantly contribute to second-lien misrepresentation.

Third, our paper contributes to the literature on mortgage fraud by proposing a preference-based explanation. A substantial body of literature documents the prevalence of mortgage fraud and its economic impact (Elul et al., 2010; Piskorski et al., 2015; Ambrose et al., 2016; Griffin and Maturana, 2016; Mian and Suf, 2017; Kruger and Maturana, 2021). While previous studies have considered associated loan characteristics and household features, the role of preference in the decision-making process remains unclear. From a cultural perspective, Conklin et al. (2022) provide empirical evidence that religiosity helps constrain fraudulent activity but do not distinguish between ethics and risk channels. Bypassing the issue of differentiating ethics and risk preference, we show that gambling preference[4] influences mortgage applications.

Fourth, our paper builds on the empirical literature linking gambling preferences with investment decisions. Prior research has examined the effect of gambling preference on the stock market (Barberis and Huang, 2008; Kumar, 2009; Kumar et al., 2011; Barberis et al., 2016, 2021), bond market (Benartzi and Thaler, 1995), and option market (Baele et al., 2019). Additionally, literature explores its impact on corporate policy decisions (Kumar et al., 2011; Chen et al., 2014) and fund strategies (Shu et al., 2012). To the best of our knowledge, we are the first to apply the concept of gambling preference to the mortgage market in the context of household investment decision-making. This is important because existing literature on behavioral biases in the housing market predominantly centers on pricing dynamics and speculative bubbles from the investment perspective (Lux, 1995; Genesove and Mayer, 2001; Farlow, 2004; Shiller, 2005; Farlow, 2013), whereas relatively little attention has been paid to behavioral influences on mortgage choices, which is the financing side of housing decisions.

The rest of the paper is organized as follows. Section 2 describes the data and variables. In Section 3, we examine the effect of local demographic characteristics on second-lien misrepresentation. In Section 4, we interpret the effect of gambling preference and assess whether it aligns with borrower behavior or lender practice. Section 5 investigates the outcomes of gambling preference-related misrepresentation. Robustness checks are conducted in Section 6, and conclusions are presented in Section 7.

[4] See Kumar (2009), Kumar et al. (2011), Kumar et al. (2016) for how to differentiate gambling preference from ethics.

2 Data and Summary Statistics

Our sample comprises three main groups of data: loan-related records, religiosity data, and demographic information, covering the period from 2005 to 2007. These data are sourced from various datasets. Loan-related records include loan-level mortgage data from BlackBox Logic (now part of Moody's) and borrower-level credit report information from Equifax. Religiosity data, which provides county-level information on prevalent religious adherence, is obtained from the American Religion Data Archive (ARDA). Demographic information at the county and ZIP code levels, such as age and income, is sourced from the U.S. Census Bureau. We also collect crimes and voter registration data from National Neighborhood Data Archive (NaNDA) and the Uniform Crime Reporting (UCR) Program Data available through the ICPSR website. Additionally, we collect house price data from the Federal Housing Finance Agency (FHFA) and unemployment data from U.S. Bureau of Labor Statistics.

2.1 Second-lien Misrepresentation Measure

Second-lien misrepresentation occurs when a first-lien loan backed by property is reported as having no associated higher lien but is actually financed with a simultaneously originated second mortgage identified by credit bureau data. This misrepresentation allows the borrower to take on additional debt, reducing their incentive to repay the loans and making the initial debt riskier. We identify second-lien misrepresentation by comparing loan-level mortgage data from BlackBox Logic with borrower-level credit report information from Equifax. Following the procedure in Piskorski et al. (2015), we first focus on first-lien loans with a merge confidence interval greater than or equal to 0.89. Second, we select loans with a new second lien originating within one month before or after the first lien[5]. This filter, using credit information, retains first-lien loans that truly have a simultaneous second lien. Third,

[5]Piskorski et al. (2015) uses a 45-day range, while Zhang et al. (2024) uses a one-month range. Since Equifax data identifies the close date of the second lien at the month level, we conservatively use a one-month range. A 45-day range can be achieved by assuming the close date is in the middle of the month. We also checked the 45-day range and found minor differences in the results.

to identify misreported second-lien loans, we require the loan to have a non-missing reported cumulative loan-to-value (CLTV) ratio within 1% of its loan-to-value (LTV) ratio. The small difference between the reported CLTV and LTV indicates that the borrower reports no simultaneous second lien.

2.2 Gambling Preference and Religiosity

Following Kumar et al. (2011), we construct county-level gambling preference based on religious data, specifically the ratio of Catholic residents to Protestant residents (*CPRATIO*). Although direct measures of local gambling preference are unavailable[6], we can infer the propensity by examining the proportion of different religious populations, which have distinct attitudes towards gambling according to their religious views. In the U.S., two widespread religions with differing views on gambling are Catholicism and Protestantism. While Protestant churches generally oppose gambling, Catholic churches maintain a tolerant attitude towards moderate levels of gambling. These differing views between the two religions are empirically supported to extend to financial markets (Kumar, 2009; Kumar et al., 2011; Han and Kumar, 2013; Chen et al., 2014). Therefore, regions with higher Catholic–Protestant ratios exhibit stronger gambling propensities.

We also construct county-level religiosity (*REL*) following Kumar et al. (2011). According to Hilary and Hui (2009), this measure of religiosity reflects risk attitudes, with higher values of REL indicating greater risk aversion. Moreover, religiosity captures social norms and ethical behavior, which play a significant role in deterring various types of mortgage misrepresentation (Conklin et al., 2022). Therefore, including the overall level of religiosity in the county ensures that our proxy for local gambling preference is independent of religion-induced risk aversion and ethical norms, thereby strengthening the potential causal interpretation of *CPRATIO*.

We use the dataset "Longitudinal Religious Congregations and Membership File, 1980-

[6]Gambling industry employment can serve as alternative measures of local gambling preference but has several limitations. We therefore use it only as robustness checks. See Appendix for further discussion.

2010 (County Level)" from ARDA to capture county-level geographical variation in religious composition. We sum the number of adherents in the religious traditional categories Evangelical Protestant, Mainline Protestant, and Black Protestant for the Protestant population and the number of adherents in the category Catholic for the Catholic population. We use the sum of adherents in all categories for the total religious adherents in the county. Since the data is available for each decade from 1980 to 2010, we follow previous studies to linearly interpolate the data to obtain values for missing years (Alesina and La Ferrara, 2000; Hilary and Hui, 2009; Kumar et al., 2011; Chen et al., 2014), and then calculate the *CPRATIO* (i.e., Catholic population to Protestant population) and *REL* (i.e., total religious adherents to total population) for each year.

2.3 Other Socioeconomic Characteristics

In addition to gambling preference and religiosity, we also examine a rich set of local socioeconomic characteristics, covering both demographic factors and local economic conditions. Specifically, we use county-level data from 2000 on education, marriage, residential area, population, age, male-female ratio, and minority share, as well as ZIP code-level income information, from the U.S. Census Bureau. We further incorporate county-level crime statistics from 2005 to 2007 and voter registration data from 2006 obtained from the ICPSR. These demographic characteristics are closely related to risk preferences, ethical norms, cultural background, and financial literacy, all of which are likely to influence household financial decisions.

To capture local economic conditions that also play a crucial role in mortgage decisions, we employ county-level monthly unemployment rates from the U.S. Bureau of Labor Statistics. In addition, we use annual ZIP code-level house price indices (HPI) to construct measures of house price appreciation.

2.4 Mortgage Microdata

In addition to the second-lien misrepresentation measure, we also obtain other loan-level mortgage data from BlackBox Logic. The database includes a comprehensive set of loan characteristics at origination, such as the loan interest rate, borrower's FICO credit score, initial loan balance, loan-to-value ratio, amortization type (e.g., full amortization, interest-only, negative amortization), income documentation type (e.g., full, low-doc, or no-doc), interest rate type (e.g., fixed or adjustable), prepayment penalty, loan purpose (e.g., purchase or refinance), reported occupancy status (e.g., owner-occupied, investment, or second-home), delinquency method (i.e., MBA or OTS), and delinquency status (e.g., current, 30, 60, 90 days, etc.). The definitions of variables constructed from this information are presented in Table 1.

2.5 Descriptive Statistics

We study the effect of borrower characteristics on second-lien misrepresentation in a sample that includes loans without simultaneous second liens, loans with correctly reported simultaneous second liens, and loans with misreported simultaneous second liens. We select the sample period of 2005 to 2007, during which mortgage misrepresentation measures can be calculated using the databases and misrepresentations are not rare. Table 2 shows the summary statistics. For each continuous variable, we report the number of observations, mean, standard deviation, 25th percentile, 50th percentile, and 75th percentile. For dummy variables, we report the number of observations and mean. The continuous variables are winsorized at the 0.5 percent level to ensure our results are not driven by extreme values.

In our sample, we have 3,031,921 loans for which the borrower's area *CPRATIO* is available[7]. Among these loans, about 7.15 percent have misreported second liens, while 13.91 percent have reported second liens, resulting in a total of 21.06 percent. This percentage is

[7]We also exclude cases where the LTV is greater than 99% since our measure is calculated using LTV. An LTV greater than 99% is likely erroneous or highly unlikely to have a simultaneous second lien.

comparable to the one reported by Griffin and Maturana (2016) (10.2%), who used a different database and included all loans (honestly reported, fraudulently hidden, and truly without second liens). Figure 1 plots the nationwide distribution of the county-level proportion of second-lien misrepresentation in all loans, showing variations both across and within states.

Our primary independent variable of interest is local gambling preference (*CPRATIO*). While there are 3,090 available U.S. counties from 2005 to 2007, we only retain county-year level data where loan-level data is also available. For the 8,716 data points, the mean *CPRATIO* is 0.51 and the median is 0.19, indicating a large positive skewness[8]. To interpret *CPRATIO* as a measure of local gambling preference, the most important related variable is religiosity (*REL*). This variable helps distinguish *CPRATIO* from interpretations based on religion-induced risk aversion, and it allows us to assess whether local religious composition is large enough to meaningfully capture local preferences. Among the 8,716 county-year observations, the tenth percentile of *REL* is about 37 percent and the median is about 59 percent, indicating that most counties have a substantial share of religious population.

Since we divide subsamples by *CPRATIO* in the paper, it is important to see whether this separation is close to the division by other local structural determinants, which may cause the separation capture similar information by other local measures. To assess this, we calculate pairwise correlations of all local structural determinant variables at the county-year level. *CPRATIO* generally exhibit low correlations with other characteristics, suggesting that it capture distinct information not encompassed by other common local measures. Moreover, since we examine a rich set of local structural determinant variables in groups, it is also important to see whether within each group, the variables have high correlation. We find that most characteristics display low to moderate correlations. As expected, *Income* and *Education* are relatively highly correlated, since individuals with higher education levels typically earn higher incomes. Similarly, *Urban* and *Total population* show a relatively strong correlation, consistent with the idea that counties in urban areas tend to be more

[8]The mean and median *CPRATIO* from 1980 to 2005 are 0.60 and 0.23, respectively, in Kumar et al. (2011), which is quite close to our results if we include all counties.

densely populated. Nonetheless, because none of these correlations are high enough to raise concerns about multicollinearity, and since each variable captures a different aspect of borrower characteristics, we include all of them in the regressions. The full correlation matrix can be found in Appendix.

3 Local Socioeconomic Characteristics and Second-lien Misrepresentation

In this section, we first investigate the relationship between second-lien misrepresentation and local socioeconomic characteristics that reflects household's feature in making financial decision. We conduct loan-level regression analysis of second-lien misrepresentation on the local socioeconomic characteristics using the following form of specification:

$$\begin{aligned}\text{Misreported second} = \;& \alpha + \beta_1 \times \text{Local socioeconomic characteristics} \\ & + \beta_2 \times \text{Loan characteristics} + \text{Fixed effects}\end{aligned} \tag{1}$$

where *Misreported second* is an indicator that equals one if the first-lien loan has simultaneous second-liens that are misreported at the loan application. *Local socioeconomic characteristics* includes the vector of all local characteristics, such as *CPRATIO*, *REL*, and *Education*. *Loan characteristics* includes the vector of all loan-level information at the origination regarding the loan characteristics and borrower characteristics, such as *Interest rate*, *Balance*, and *FICO*. *Fixed effects* include originator, state, and half-year fixed effects, each of which may influence mortgage misrepresentation[9]. Because most local characteristic variables are measured at the county level, we cluster heteroskedasticity-robust standard errors by county in all specifications. All continuous variables are winsorized at the 0.5 percent level. Additionally, we standardized all the continuous variables so that the coefficient magnitude

[9] Although both originators and underwriters play important roles (Griffin and Maturana, 2016) and the related fixed effects are used in different literature, we control for originator fixed effects because originators are the agency that interact with the borrowers so that they may affect the decision of borrowers while underwriters do not have such influence in the borrowers decision-making process.

is comparable. Following Piskorski et al. (2015) and Yavas and Zhu (2024), we use OLS regressions in the main tests despite the binary nature of the dependent variable, and rely on probit models and causal forest approaches as robustness checks.[10]

Table 3 reports the regression results. We introduce the local socioeconomic characteristics by groups separately into the regression and then include them all to study their effects. Column (1) looks at social environment. Column (2) tests economic status. Column (3) examines local market conditions. Column (4) studies household structure. Column (5) includes all variables.

We now examine each variable in detail, focusing on their interpretation once other controls are included and evaluating their robustness in explaining second-lien misrepresentation. First, for the social environment, we consider *CPRATIO*, *REL*, *Crime rate*, and *Republic*. These variables capture different aspects of local social composition and may shape households' leverage choices by influencing local attitudes toward risk-taking and compliance with rules. *CPRATIO* primarily serves as a proxy for gambling preference once common local demographics are controlled for, and it typically affects financial decisions when payoffs are positively skewed (Kumar et al., 2011). It is consistently positive and significant in all specifications, indicating that counties with stronger gambling tendencies exhibit higher misrepresentation rates. *REL* reflects religion-induced risk aversion and ethical norms (Hilary and Hui, 2009), which are relevant for certain types of mortgage misrepresentation (Conklin et al., 2022). Its coefficients are insignificant in both columns (1) and (5), suggesting that religion-based risk attitudes and ethical norms do not meaningfully affect households' propensity to take excessive leverage. *Crime rate*, an important determinant of house prices and credit-market conditions (Mian and Sufi, 2009), is used as a proxy for moral norms or the local illegitimate opportunity structure (Dyck and Zingales, 2004; Baumer et al., 2017). It is positively correlated with misrepresentation, implying that weaker moral norms may in-

[10]The OLS model has an important advantage in allowing the inclusion of multiple fixed effects. By contrast, probit or logit models generally yield inconsistent estimates when more than one fixed effect is included in large samples (Wooldridge, 2010).

crease the likelihood of excessive leverage through misreporting. *Republic* captures partisan leaning toward Republic, and partisan bias can shape households' economic expectations and spending when their affiliated party controls the White House (Mian et al., 2023). It loads positively but insignificantly, suggesting that when Republicans control the White House[11], Republican-leaning households may be marginally more aggressive in borrowing, although the evidence is weak.

Second, for economic status, we examine *Education* and *Income*. These factors may influence leverage decisions by shaping both financial capacity and households' ability to process financial information. *Education* is typically positively associated with financial literacy (Campbell, 2006). It shows a positive but unstable association with misrepresentation, consistent with better-educated households being more capable of exploiting financial opportunities. *Income* reflects household financial strength and is a key determinant of consumption and borrowing (Campbell, 2006). It loads negatively across specifications, suggesting that stronger financial conditions reduce the need to take excessive leverage through misreporting.

Third, for local market conditions, we consider *Unemployment*, *HPA*, *Urban*, and *Total population*. These variables may affect leverage choices by altering both the economic environment in which borrowing occurs and the opportunities or constraints households face. *Unemployment*, which captures local labor market distress, may influence leverage decisions through income uncertainty (Corradin, 2014). Its coefficients are negative but insignificant, implying that joblessness may be associated with slightly lower excessive leverage, though the effect is limited. *HPA*, reflecting past house price appreciation, shapes expectations of future price growth and typically boosts willingness to take on leverage (Adelino et al., 2016). It remains significantly positive, indicating that strong recent house price dynamics induce households to borrow more aggressively to capture potential future gains. *Urban*, capturing settlement patterns, may affect misrepresentation through differences in credit access

[11] During our sample period (2005–2007), the Republican Party controlled the White House.

(Rogers, 2006). It is negative and becomes significant with richer controls, implying that rural areas exhibit greater misrepresentation. *Total population*, reflecting county size, may shape local housing and credit markets by influencing households' demand for loans and by affecting the development and sophistication of local financial markets. It is significantly positive across all specifications, suggesting that more populous counties tend to experience higher misrepresentation rates.

Last, for household structure, we include *Married*, *Over65*, *Male–female ratio*, and *Minority*. These variables may shape leverage choices by influencing financial needs, risk preferences, and borrowing constraints. *Married*, which reflects household stability and homeownership propensity (Grinstein-Weiss et al., 2011), may increase leverage demand while reducing risk-taking. Its small and insignificant coefficients imply little net effect. *Over65*, the share of older residents who are typically risk-averse but borrowing-constrained (Barsky et al., 1997; Nakajima and Telyukova, 2017), may affect excessive leverage through lower risk tolerance or lower income. It has positive coefficients that become more significant with richer controls, indicating that older households may also participate more in misrepresentation.[12] *Male–female ratio*, which reflects gender-based risk attitudes (Croson and Gneezy, 2009), is negative but insignificant, suggesting that gender composition plays a minor role. *Minority*, typically associated with higher-cost mortgage terms and lower income (Bayer et al., 2018; Kennickell, 2009), loads positively, indicating that structural disadvantages faced by minority borrowers may push them toward more aggressive borrowing through misreporting.

Taken together, these results show that among all local socioeconomic characteristics that could plausibly influence excessive leverage through mortgage misrepresentation, only *CPRATIO*, *HPA*, and *Total population* emerge as both robust and substantial in magnitude. While *HPA* and *Total population* are standard local market conditions known to affect household leverage decisions, the striking result is that *CPRATIO*, which reflects borrowers' be-

[12]This may contradict common expectations given high homeownership rates and greater risk aversion among older households. One potential explanation is that older households may have stronger motives to reduce financial burdens or extract home equity to support current spending. In subsample tests, the effect of *Over65* appears only in refinance and primary-residence subsamples, which is consistent this explanation.

havioral traits, also plays an important role in explaining excessive leverage. A one–standard deviation increase in *CPRATIO* raises the probability of second-lien misrepresentation by 0.14 percentage points, even after controlling for other local characteristics, detailed loan and borrower fundamentals, and fixed effects.[13] This effect is comparable to borrower-level fundamentals such as the interest rate (–0.72 percentage points) and the FICO score (–0.62 percentage points). By contrast, other demographics, such as education, religion, or income, are not robust across specifications or display much smaller coefficients. These findings motivate our focus on *CPRATIO*, which we interpret as a proxy for local gambling preference, a distinct behavioral trait that systematically drives mortgage misrepresentation in a way that conventional demographics do not.

4 Interpreting the Effect of Gambling Preference: Borrower Behavior or Lender Practice

From the previous section, we find that second-lien misrepresentation is more likely to occur in counties with higher values of *CPRATIO*. We interpret *CPRATIO* as a measure of local gambling preference, following Kumar et al. (2011), who show that regions with higher *CPRATIO* exhibit stronger gambling and speculative tendencies. Gambling preference, in this context, refers to a willingness to accept a high probability of small losses for a small chance of a large gain (Markowitz, 1952). This behavioral tendency can influence financial decisions by making households more inclined to pursue opportunities with large potential payoffs despite substantial risks. In our setting, it implies that in areas with stronger gambling preferences, borrowers are more willing to take on excessive leverage through second-lien misrepresentation.

From the household perspective, misreporting a second lien creates an opportunity with

[13] The absolute magnitude appears small because the unconditional misrepresentation rate is only 7.15 percent, but the effect represents a roughly 2 percent increase relative to the baseline rate.

asymmetric payoffs. The upside combines access to homeownership and the possibility of obtaining better loan terms. Homeownership carries both emotional and financial appeal. Emotionally, it is associated with the American Dream (Clinton, 1995; Bush, 2003), while financially, it offers the potential for wealth accumulation through house price appreciation and mortgage-interest tax deductions (Bostic and Lee, 2008; Goodman and Mayer, 2018). During the housing boom before the 2008 financial crisis, this upside appeared especially attractive because most households and lenders believed that house prices would continue to rise, which made the expected payoff distribution strongly right-skewed. Even if prices rose only moderately or stagnated, borrowers still expected to preserve some gains or at least maintain ownership.[14] By contrast, the downside was relatively limited. The most common outcome of detection was loan denial, which entailed small application costs and minimal switching effort in an environment with abundant credit supply. Before the loan closed, lenders could file Suspicious Activity Reports, which sometimes led to investigation and charges. After closing, borrowers might face fines, restitution, or probation if misrepresentation was uncovered (Henning, 2009). In practice, however, law enforcement during this period focused on fraud for profit rather than fraud for housing (FBI, 2006, 2007). As a result, households who misreported to obtain housing rarely faced severe penalties. The asymmetry between large perceived gains and modest expected losses produces a right-skewed payoff structure that is particularly appealing to borrowers in areas with stronger gambling preferences.

However, the measure of gambling preference is constructed at the local level rather than the borrower level, which raises the possibility that the effect on second-lien misrepresentation may operate through lenders' practices rather than borrower decisions. Prior research has shown that both originators and underwriters can play significant roles in mortgage misreporting (Griffin and Maturana, 2016; Piskorski et al., 2015). These intermediaries were

[14] A decline in house prices may create financial losses, but borrowers who continue making payments can keep their homes. If they choose to default because of negative equity, the loss they incur consists of the down payment and the portion of principal they have already repaid. Since high leverage requires only a small initial equity stake, this total loss remains relatively small for borrowers who took on excessive leverage.

often aware of hidden second liens yet still allowed them to be misreported. More recently, by comparing portfolio loans with privately securitized mortgages, Yavas and Zhu (2024) provide evidence that second-lien misrepresentation tends to arise in the early stages of intermediation by lenders rather than at the underwriting stage.

Therefore, it is necessary to assess whether the observed relationship should be attributed primarily to borrower behavior or to lender practices. In the remainder of this section, we examine evidence from both perspectives and show that the patterns are more consistent with an interpretation based on borrower behavior.

4.1 Evidence from Borrower Behavior

According to Christensen et al. (2018), the impact of gambling preference on misreporting should be more pronounced in situations where individuals face stronger motives. Since our interpretation primarily emphasizes the borrower's perspective, we examine whether the effect is amplified when borrowers' motives are greater, using subsample analysis.

We begin by examining subsamples defined by loan-level characteristics that directly capture borrower motives. Specifically, we partition the sample by the loan's property occupancy type (primary or non-primary) and by the loan's purpose type (purchase or refinance). A primary occupancy type indicates that the property is owner-occupied, which is tied to the borrower's need for stable housing and the utility derived from living in the home. In contrast, a non-primary occupancy type implies that the property is used as a second home or investment, suggesting that the borrower already owns a residence and is motivated by discretionary consumption or wealth accumulation. Because the motive to misrepresent should be greater when stable housing is at stake, we expect the effect of gambling preference is more evident in the primary occupancy subsample. Similarly, if the loan purpose is a purchase, the borrower does not yet own the property and must obtain it to begin consumption or investment. By contrast, if the purpose is refinance, the borrower already owns the property and seeks to improve financial conditions by lowering costs or accessing home equity. In this case, the motive to misrepresent should be stronger in purchase loans, making them a setting where gambling preference is expected to play a larger role.

Table 4 reports the subsample results. Columns (1) and (2) present estimates for the primary and non-primary groups. Most observations in the overall sample fall into the primary category. Moreover, only the coefficient of *CPRATIO* in the primary subsample is significant, and its economic magnitude is substantially larger than that in the non-primary group. This finding suggests that borrowers with stronger gambling preferences are more likely to misrepresent second liens when the property is intended as a primary residence, consistent with our expectation that the relationship between gambling preference and misrepresentation is stronger in the primary subsample, where borrower motives are greater.

Columns (3) and (4) present the results for the purchase and refinance subsamples. About 43 percent of the observations come from the purchase group. The coefficient of *CPRATIO* in the purchase subsample is significant at the 1 percent level, whereas it is only marginally significant in the refinance subsample. Additionally, the economic magnitude in the purchase group is much larger than in the refinance group. This indicates that borrowers with stronger gambling preferences tend to misrepresent second liens primarily when purchasing a house, reinforcing our expectation that the relationship between gambling preference and misrepresentation is stronger in the purchase subsample, where borrower motives are greater.

We also conduct subsample analyses based on characteristics that not only relate to broader debates about the origins of the financial crisis but also capture variation in borrowers' motives. Specifically, we partition the sample by area income (high or low *Income*), credit score (high or low *FICO*), and past house price appreciation (high or low *HPA*). The high- and low-income subsamples are defined by comparing local household median income with the sample median at the ZIP code level. The high- and low-credit-score subsamples are divided using a FICO cutoff of 670[15]. The high- and low-HPA subsamples are defined

[15]670 is the boundary between fair and good levels as evaluated by the institution. We also tried the median of the sample, 682, which yielded similar results.

based on the state-year median *HPA*, calculated from all available ZIP code-level data.

Area income and credit score are closely tied to the credit expansion argument, which centers on whether the expansion of mortgage credit leading up to the crisis was widespread across the population. In our setting, gambling preference should affect all groups, as all contributed to the crisis. However, low-income and low-credit-score borrowers are expected to show stronger effects because they have greater incentives to misrepresent, stemming from tighter financial constraints (low income) or more limited credit access (low FICO). As shown in Table 5, columns (1) to (4), the economic magnitude of *CPRATIO* is large across different groups[16], and the effects are larger and more significant in the low-income and low-FICO subsamples. This pattern is consistent with the credit expansion view and also reinforces the interpretation of borrower gambling preference.

Past house price appreciation is closely related to the belief distortion channel, which emphasizes that households during this period held overly optimistic expectations of future price growth. In our context, households in areas with higher past appreciation are more likely to extrapolate those increases into the future (Shiller, 2007), creating stronger incentives to misrepresent due to higher expected returns. As shown in Table 6, columns (5) and (6), the coefficient of *CPRATIO* is larger in magnitude and more statistically significant in high-HPA areas. This evidence indicates that our findings are also consistent with the belief distortion channel, while continuing to support the interpretation of borrower gambling preference.

Overall, the subsample analysis shows that second-lien misrepresentation rises more sharply with gambling preference when borrowers face stronger motives. This evidence support our interpretation of the effect from borrower's perspective.

[16]Recall that in the baseline results, the coefficient of CPRATIO is about 0.0014.

4.2 Evidence from Lender Practice

To assess this issue from the perspective of lenders' practices, we employ the setting of ease of securitization around a FICO score of 620, as described by Keys et al. (2010). Due to the underwriting guidelines established by government-sponsored enterprises, Fannie Mae and Freddie Mac, low documentation loans made to borrowers with FICO scores of 620 or higher are much easier to securitize. Consequently, lenders have lax screening standards for such loans, and a regression discontinuity design could reveal an upward jump in the default rate for low documentation loans. Additionally, Griffin and Maturana (2016) find that the second-lien misrepresentation rate also jumps at this cutoff, implying that this type of misrepresentation is associated with lenders' incentives to securitize the loan.

Utilizing this shock to lenders, we first study the extent to which lenders facilitate second-lien misrepresentation by comparing loans with misreported simultaneous seconds to other types of loans. Specifically, we examine the jumps in the number of loans and the jumps in the default rate for all loans, loans with correctly reported simultaneous seconds, and loans with misreported simultaneous seconds.[17] If lenders specifically facilitate second-lien misrepresentation, then the jump ratio in the number of loans with misreported simultaneous seconds should be greater than for other types of loans. Additionally, if such special facilitation of misrepresentation stems from lenders' screening efforts, the jumps in the default rate for loans with misreported simultaneous seconds should be significantly greater than those for loans with correctly reported simultaneous seconds. In contrast, if the jump ratio in the number of loans with misreported simultaneous seconds is similar to or smaller than for other types of loans, lenders do not specifically facilitate second-lien misrepresentation but increase it with the increase of other loans. Finally, if the jumps in the

[17] The misrepresentation rate is an alternative measure of the extent to which lenders facilitate misrepresentation, and we report these results in the robustness section. We choose to use the number of loans in this section because it directly reflects how the ease of securitization affects each loan type. A rate measure contains both a numerator and a denominator, so changes in the rate may arise from movements in misrepresented loans or from movements in all loans. The count-based measure avoids this ambiguity and makes it clear where the difference originates.

default rate for loans with misreported simultaneous seconds are no larger than those for loans with correctly reported seconds, this would imply that lenders' screening effort does not play a distinct role in facilitating misrepresentation.

Building on this general picture, we next investigate whether the effect of gambling preference on second-lien misrepresentation aligns with lender practices. Since the variation in second-lien misrepresentation related to gambling preference could come from either lenders or borrowers, and the shock we employed is only for lenders, if the increase in second-lien misrepresentation in high gambling preference areas is smaller, it indicates that lenders play a smaller role in increasing misrepresentation in high gambling preference areas. This implies that the positive relationship between gambling preference and second-lien misrepresentation is mainly attributed to borrowers. Specifically, we examine the jump ratio in the number of loans and the jumps in the default rate to understand how lender's roles differ across areas. If the effect of local gambling preference on second-lien misrepresentation is in line with lenders' practice (i.e., lenders make second-lien misrepresentation more prevalent in high gambling preference areas), this shock to lenders would lead to different outcomes between high and low gambling preference areas, favoring high gambling preference areas (i.e., a greater jump ratio in the number of loans with misrepresentation and a larger increase in the default rate). In contrast, a similar or smaller jump ratio in the number of misreported loans and the default rates of misreported loans in high gambling preference areas indicates that lenders did not facilitate more misrepresentation in such areas.

We take RDD approach using local linear regressions. The credit scores are normalized as follows:

$$\mathrm{C} = \mathrm{Credit\ Score} - \mathrm{Threshold} \tag{2}$$

where $Threshold$ is 620 for low documentation loans. Then, we define D to distinguish credit scores that are over or below the threshold as follows:

$$D = \begin{cases} 1, & if\ \mathrm{C} \geq 0 \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

To differentiate the effect between high and low local gambling preference areas, we define high local gambling preference ares:

$$G = \begin{cases} 1, & if\ \mathrm{CPRATIO} \geq 1.2921 \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

We use 1.2921 as the cutoff for two reasons[18]. First, only 10 percent of county-years in our sample have a CPRATIO greater than or equal to 1.2921, indicating that areas with such a CPRATIO indeed have a high local gambling preference. Second, about half of all loans in our sample are originated in these areas, making our findings about high and low gambling preference areas comparable.

To apply the RDD approach using local linear regressions, we choose a bandwidth of 10, which corresponds to a FICO score range of 610 to 630. We selected this window because it ensures that no other jumps are found in the literature[19] and it is close to the optimal bandwidth generated by a data-driven bandwidth selection method[20] (Calonico et al., 2020). We use the fixed bandwidth rather than the optimal bandwidth so that the RDD results in different cases are comparable[21]. For the default rate and number of loans in high and low gambling preference areas, we estimate the four cases separately using the following specification:

$$Y = \alpha + \beta D + \gamma C + \delta D \times C + \epsilon \tag{5}$$

where Y is the default dummy variable for loan i originated at time t or the number of loans

[18] We also tried other reasonable cutoffs, such as 1, and the results are robust.

[19] For example, FICO scores of 600 and 660 could be other potential cutoffs for jumps.

[20] The optimal bandwidth for the number of all loans is 10.50.

[21] Using the optimal bandwidth does not affect our findings.

at each FICO score. In our baseline model, we use uniform kernel for estimation.

Finally, to calculate the difference in the discontinuities (diff-in-disc) between high and low gambling preference areas, we follow Dickert-Conlin and Elder (2010) and Grembi et al. (2016) to estimate the following model:

$$Y = \beta_0 + \beta_1 D + \beta_2 C + \beta_3 D \times C + \beta_4 G + \beta_5 G \times C + \beta_6 G \times D + \beta_7 G \times C \times D + \epsilon \quad (6)$$

where β_6 is the parameter that measures the difference in the discontinuity in Y at credit score cutoff for high versus low gambling preference areas. When we apply this equation to calculate the difference in jump ratios between high and low gambling preference areas, we first rescale the data to obtain the t-statistics. We divide the data in high gambling preference areas by the estimated value at FICO 620^- in those areas. Similarly, we divide the data in low gambling preference areas by their corresponding estimated value at FICO 620^-. After this rescaling, the discontinuity estimated in high or low gambling preference areas can be directly interpreted as a multiple of its corresponding estimated value at FICO 620^-. The difference between the two discontinuities represents the difference in jump ratios between high and low gambling preference areas.

We first look at the jumps in the number of loans and the jumps in the default rate in all areas, examining the pattern of second-lien misrepresentation given a shock to lenders. Panel A of Table 7 presents the results for jumps in the total number of loans, loans with correctly reported simultaneous seconds, and loans with misreported simultaneous seconds. Overall, the total number of low-documentation loans doubles from 620^- to 620^+, consistent with the findings in Keys et al. (2010). Loans with simultaneous seconds increase more than those without, whether correctly reported or misreported. This pattern indicates not only that the ease of securitization applies to these loans but also that lenders actively facilitate such loans with the intention of securitizing them.

Panel B of Table 7 reports the jumps in default rates for all loans, for loans with correctly reported simultaneous seconds, and for loans with misreported simultaneous seconds. Consistent with previous literature, there is a jump of 0.039 in the default rate for all loans, indicating overall reduced screening for 620^+ loans. This lax screening is also observed in loans with correctly reported simultaneous seconds, with a statistically significant jump of 0.045. For the loans with misreported simultaneous seconds, the jump in the default rate is similar in magnitude but statistically insignificant, suggesting that the lax screening effect is not as pronounced as for correctly reported seconds. These findings illustrate that lenders facilitate loans with simultaneous second liens, whether correctly reported or misreported, and that the lax screening effect is no larger for misreported seconds than for correctly reported seconds.

Having examined the jumps in all areas, we next compare high and low gambling preference areas to determine whether the effect aligns with lender practices. Figure 2 compares the jump ratios in the number of loans with misreported or correctly reported simultaneous seconds across high and low gambling preference areas. For misreported seconds, the jump ratio in high gambling preference areas is noticeably smaller than in low gambling preference areas, which suggests that lenders in high gambling preference areas do not facilitate misreported loans more aggressively at the cutoff. For loans with correctly reported simultaneous seconds, the jump ratios in high and low gambling preference areas are broadly similar, with high gambling preference areas exhibiting only a slightly larger increase. Moreover, Panel A of Table 8 also shows that the increases in the total number of loans are similar in magnitude across the two groups, which indicates that the ease of securitization does not differ systematically between high and low gambling preference areas. Taken together, these findings suggest that lenders do not play a larger role in increasing second-lien misrepresentation in high gambling preference areas.[22]

[22] If borrowers contribute more to misrepresentation in high gambling preference areas, the effect of lenders on increasing the number of misreported loans is more limited, which leads to a smaller jump. In contrast, if borrowers contribute less to misrepresentation in low gambling preference areas, the effect of lenders on increasing the number of misreported loans is less constrained, which leads to a larger jump.

Figure 3 further compares the jumps in default rates for loans with misreported and correctly reported simultaneous seconds across high and low gambling preference areas. Both types of loans show smaller jumps in high gambling preference areas. When control variables are included, Panel B of Table 8 shows that the differences between high and low gambling preference areas are insignificant, a pattern that also holds for all loans. This suggests that lenders in high gambling preference areas are not more likely to facilitate misrepresentation through laxer screening. Combined with the evidence from the number of loans, these findings show that lenders in high gambling preference areas are not more inclined to increase misrepresentation than lenders in low gambling preference areas. In other words, the evidence does not support the interpretation that the gambling preference effect operates through lenders' practices.

5 Outcomes of Misrepresentation Across Gambling Preference Levels

In the previous section, we found a significant correlation between local gambling preference and mortgage misrepresentation. Our next step is to investigate how local gambling preference shapes the consequences of misrepresentation. Prior studies show that mortgage misrepresentation generally leads to worse loan performance (Piskorski et al., 2015; Griffin and Maturana, 2016), but it remains unclear whether misrepresentations arising in high-gambling-preference environments have systematically different consequences. Since gambling is typically linked to adverse financial outcomes, we conjecture that high local gambling preference may amplify the performance deterioration of loans with misreported seconds. To evaluate this possibility, we estimate loan-level regressions of delinquency on indicators for misreported simultaneous second liens, and we compare the estimated effects across high and low gambling preference areas defined in Section 4.2 using the following specification:

$$\begin{aligned}\text{Default} = \quad & \alpha + \beta_1 \times \text{Misreported second} + \beta_2 \times \text{Reported second} \\ & + \beta_3 \times \text{Controls} + \text{Fixed effects}\end{aligned} \tag{7}$$

where *Default* is an indicator that equals one if the loan becomes 90 days or more delinquent using MBA method. *Misreported second* is an indicator that equals one if the first-lien loan has simultaneous second liens that are misreported at the loan application. *Reported second* is an indicator that equals one if the first-lien loan has simultaneous second liens that are correctly reported at the loan application. *Controls* includes the vector of all other variables used in the baseline findings, such as local characteristics and loan-level information. *Fixed effects* include originator, state, and half-year fixed effects. We include control variables, fixed effects, and standard error clustering in all specifications.

The results are reported in Table 9. Column (1) reports estimates for the subsample of high gambling preference areas, and column (2) reports estimates for the subsample of low gambling preference areas. In both subsamples, loans with misreported simultaneous seconds default at significantly higher rates than loans without second liens, consistent with the general finding that misreported loans perform worse. More importantly, the estimated default penalty for misreported seconds is larger in high gambling preference areas than in low gambling preference areas. This pattern indicates that the adverse performance consequences of misreported seconds are amplified in environments with stronger gambling preference. Loans with correctly reported seconds also default more frequently than loans without second liens, and the magnitude of this effect is likewise larger in high gambling preference areas. While our focus is on misreported seconds as the concealed and more aggressive form of leverage taking, the results collectively show that high local gambling preference exacerbates the fragility of high-leverage borrowing more generally.

To gauge the economic magnitude, we compare the estimated default effects for misreported seconds across high and low gambling preference areas. In low gambling preference areas, misreported simultaneous seconds are associated with a 10.0 percentage-point higher default probability, while in high gambling preference areas the corresponding increase is 14.1 percentage points. The difference of 4.1 percentage points represents the additional default penalty attributable to the gambling environment. Multiplying this differential by the 93,925 misreported loans originated in high gambling preference areas implies roughly 3,852 additional defaults. At an average first-lien balance of about $325,000, this translates into approximately $1.25 billion in incremental first-lien exposure. This back-of-the-envelope calculation highlights that the amplification of misreported-loan risk in gambling-prone areas is economically meaningful.

6 Robustness

6.1 Nonlinear Model

The results in the baseline model are estimated using a linear probability model (OLS). To ensure our findings are not driven by this modeling choice, we also use a nonlinear specification (probit) for inference. We present the marginal effects in Table 10. Since we control for simultaneous second liens in the regression, the probit model drops all observations without simultaneous seconds due to perfect failure prediction of misrepresentation. The marginal effects of *CPRATIO* remain significant while other demographics related to household risk preference, ethics, culture, and financial literacy continue to show little robustness.

A potential concern with this approach is that the sample is limited to loans with simultaneous seconds, so the estimated misrepresentation rate no longer reflects the entire loan population. To address this issue, we re-estimate the probit model without the loan-level indicator for simultaneous seconds. Instead, we control for the prevalence of simultaneous seconds at the county level. The marginal effects from this specification are reported in Table 11. Once again, *CPRATIO* remains robustly significant, while the significance of other demographics changes somewhat depending on the modeling choice.

6.2 Causal Forest and Causality

To enhance the causal inference between gambling preference and mortgage misrepresentation and to explore the heterogeneous treatment effects of mortgage misrepresentation on mortgage default conditioned on gambling preference, we employ the causal forest approach proposed by Wager and Athey (2018). This method uses an augmented inverse propensity-weighted estimator with the random forest method in machine learning, incorporating an honesty condition. It provides double robustness (compared to propensity score matching) and high efficiency for high-dimensional models (compared to nearest-neighbor matching) in observational settings[23]. This approach has also been used in finance, such as in Rampini and Viswanathan (2022) for secured debt and Gulen et al. (2021) for corporate finance, and has shown better performance than traditional causal inference designs (Gulen et al., 2021) in terms of accuracy.

We first use this approach to investigate the causality between gambling preference and mortgage misrepresentation. In the causal forest approach, we set mortgage misrepresentation as the outcome and *CPRATIO* as the treatment. We use all control variables from Table 3 as matching variables to help grow trees and forests[24]. All continuous variables, including *CPRATIO*, are winsorized at the 0.5 percent level and then standardized. Since fixed effects are not applicable in this approach, we create a variable for half-year periods, starting from the first half of 2005 as 1, to account for potential time effects. Similar to Athey and Wager (2019), who cluster observations by school ID, we cluster observations by state but give each unit the same weight so that larger clusters receive more weight[25]. By using such cross-fitting, our results do not solely come from any single state, but states with a greater number of observations do have greater weights. To balance computational

[23]See Wager and Athey (2018) for statistical illustration. See Athey and Wager (2019) for an example application.

[24]We require observations to be non-missing for all variables.

[25]Due to the limitation of the grf package in R, which requires the matching variables to be numerical, we do not create a numerical variable for originators to avoid falsely treating closer values as shorter distances. We also do not cluster observations by originator because the large number of originators would excessively increase the computational burden.

capacity and the accuracy of confidence intervals, we grow 2000 trees for the forest, which is also the default setting in Athey and Wager (2019). For the honesty property, we use the default splitting fraction: for each sample, we use 50 percent of the data for splitting and the remaining data for estimation. Table 12 shows the results using the causal forest. The coefficients in the regressions of second-lien misrepresentation remain significant, and the magnitude is similar to those estimated by OLS.

Second, we study the heterogeneous treatment effects of mortgage misrepresentation on default conditioned on different levels of gambling preference. We set default as the outcome and mortgage misrepresentation as the treatment. Except for adding *CPRATIO* to the matching variable matrix, the other matching variables and clustering variables are the same as those in the above causal forest regressions. For the same considerations, we also grow 2000 trees for the forest and use 50 percent of the data in each sample for splits. After estimating the causal forest, the average treatment effects of mortgage misrepresentation on default are calculated for each quarter of the sample sorted by *CPRATIO*. Figure 4 shows the results for second-lien misrepresentation. The treatment effects are generally larger when *CPRATIO* is greater, which is consistent with the results from the OLS model.

6.3 Alternative Gambling Preference Measures

While *CPRATIO* serves as our primary measure of gambling preference, we also employ alternative measures to address concerns that *CPRATIO* may reflect broader religious differences rather than gambling-specific tendencies. Following studies such as Markham et al. (2023), which use the density of gambling outlets as a measure of gambling exposure, we incorporate employment of gambling establishments as alternative measures that reflect gambling preference through local supply. Specifically, we utilize per capita employment in the gambling industry at the core-based statistical area (CBSA) level. This alternative helps isolate local gambling behavior more directly. We regress the second-lien misrepresentation dummy variable on the alternative measure, alongside other local controls at the CBSA level, loan-level controls, and fixed effects (i.e., state, lender, and half-year). See Appendix for detailed descriptions of the data and empirical tests.

Another alternative measure is to incorporate the shares of Latter-Day Saint (Mormons) and Jewish (Jews) populations into the religious composition used to construct *CPRATIO*, following Kumar et al. (2011). Individuals of Jewish faith tend to behave similarly to Catholics in their acceptance of gambling activities, whereas the gambling attitudes of Latter-Day Saints are more closely aligned with those of Protestants. Including the Jewish population in the numerator (Catholic) and the Latter-Day Saint population in the denominator (Protestant) therefore continues to capture attitudes toward gambling while alleviating concerns that the measure reflects other specific differences between Catholic and Protestant populations. As in the baseline regressions, we regress the second-lien misrepresentation indicator on this adjusted measure along with the full set of local controls at either the county or ZIP code level.

Table 13 presents the results of the primary regressions. Column (1) reports the specifications using per capita employment in the gambling industry, and Column (2) presents the specifications using *CPRATIO* adjusted to incorporate the shares of Mormons and Jews. Both alternative measures yield significantly positive coefficients, which suggests a strong association between local gambling preference and the likelihood of second-lien misrepresentation. Overall, the evidence based on per capita gambling employment and the adjusted *CPRATIO* supports the interpretation that second-lien misrepresentation is more prevalent in areas with stronger gambling preference.

6.4 RDD and Diff-in-disc concerns

Since the purpose of RDD is to show that lenders do not specifically facilitate second-lien misrepresentation compared to loans with simultaneous seconds in general, we need to verify that the underlying borrower and collateral fundamentals are smooth at the cutoff. Continuity in these pre-determined variables ensures that borrowers above and below the cutoff are drawn from comparable populations, so that differences in the probability of having a second lien or misreporting one can be attributed to changes in securitization incentives rather than to shifts in borrower quality[26]. Table 14 presents the results, and all local characteristics, such as *CPRATIO* and *REL*, and collateral fundamentals, such as *LTV* and *Balance*, are smooth around the cutoff.

Our main evidence regarding the interpretation from lenders' practices is from the the comparison of high and low gambling preference. To validate our application of the diff-in-disc approach in comparing loan volumes and default rates between high and low gambling preference areas, we address several methodological concerns. First, we examine and test the three core assumptions outlined by Grembi et al. (2016) regarding the implementation of the diff-in-disc framework. Second, we assess the sensitivity of our results to key modeling choices, including the inclusion of control variables, bandwidth selection, and estimation kernel specification. Our findings remain robust across these variations. Third, we perform a placebo test to confirm that the observed effects are not driven by random variation. For full details of these robustness checks and supporting evidence, see Appendix. Overall, these tests confirm the validation of the diff-in-disc approach and provide evidence that does not support the interpretation of the gambling preference effect through lenders' practices.

6.5 Misrepresentation Rate

When exploring the effect of gambling preference from the perspective of lenders' practices, we use the jumps in the number of loans. The advantage of using this measure compared to the misrepresentation rate is that it allows us to study changes in different types of loans

[26]This approach differs from Keys et al. (2010), who use the 620 cutoff to establish a causal link between ease of securitization and lenders' screening effort, and from Yavas and Zhu (2024), who argue that lenders' misreporting itself was a direct consequence of securitization. Our purpose is instead to benchmark misreporting against the facilitation of second-lien origination, showing that both margins respond similarly to securitization incentives. In this setting, the relevant validity check is continuity in borrower and collateral fundamentals, while discontinuities in contract terms, such as interest rates, ARM versus FRM incidence, prepayment penalties, negative amortization features, cash-out refinancing, or occupancy type, reflect lender responses to securitization demand and should be interpreted as part of the treatment channel rather than as threats to the RDD design.

without the complication of shifts in both the numerator and denominator. However, since the number of loans is aggregated, one concern is that this approach does not employ control variables as in the baseline regression, making the results subject to differences in controls. To address this issue, we also examine the jumps in the rate of having correctly reported seconds and the rate of having misreported seconds, estimated with the same control variables as in the baseline regressions.[27] Table 15 presents the results for the raw rate, the rate after controlling for all other variables, and the rate after controlling for all other variables and fixed effects. Reported seconds are similar between high and low gambling preference areas, with the high areas showing a slightly greater jump. In contrast, misreported seconds are smaller in high areas, although the difference may not be statistically significant once all controls and fixed effects are included. These results provide evidence that does not support the interpretation based on lenders' practices, under which the jump in misrepresentation should be greater in high gambling preference areas.

6.6 Alternative Measures of Default

Borrowers can default on their mortgages for various reasons over different time horizons. Our measure of default uses the MBA method of delinquency for 90 days or more and is restricted to the first three years after origination. By adjusting the definition to be more restricted (fewer default cases) or less restricted (more default cases), we can investigate whether the effect found in section 5 is generally applicable. Therefore, we consider the following default measures: 90 days or more delinquency using the MBA method in the first two years after origination (more restricted), 60 days or more delinquency using the MBA method in the first three years after origination (less restricted), and bankruptcy/foreclosure/REO in the first three years after origination (more restricted).

Table 16 presents the results of different default measures. The significance of second-lien misrepresentation is robust across all definitions of default, and in every specification

[27] We do not calculate the rate at the county level. Instead, we use the dummy of second-lien misrepresentation directly so that we can include loan-level controls as in the baseline model.

the coefficient on *CPRATIO* is larger in high gambling preference areas. Additionally, when the default measure is less restricted, the effect of mortgage misrepresentation is greater. The effect of gambling preference-related mortgage misrepresentation is also greater when the default measure is less restricted.

7 Conclusion

High household leverage was an important feature of the housing boom preceding the financial crisis. A particularly striking form of excessive leverage taking was the misrepresentation of second liens, which allowed borrowers to overstate their borrowing capacity and obtain mortgages beyond sustainable limits. This paper examines how borrower heterogeneity shaped the decision to take on such high leverage.

Using a religion-based proxy for gambling attitudes, we find that second-lien misrepresentation is more prevalent in gambling-prone counties. The relationship is especially strong when borrower incentives to misreport are high. In contrast, evidence from the FICO 620 securitization cutoff provides little support for lender-based explanations. We also show that the negative performance consequences of misrepresentation are amplified in high-gambling-preference areas. Taken together, the results suggest that borrower behavioral traits, specifically gambling preference, played a significant role in fueling excessive leverage before the crisis. Recognizing the influence of such traits is important for understanding the fragility of mortgage markets and the behavioral foundations of financial instability.

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Table 1
Variable Definitions

Variable name	Definition
Mortgage Reporting	
Misreported second	Indicator that equals one if the borrower has simultaneous second liens and misreported the second liens on the loan application
Reported second	Indicator that equals one if the borrower has simultaneous second liens and correctly reported the second-liens on the loan application
Simul. second	Indicator that equals one if the borrower have simultaneous second liens regardless the reporting status on the loan application
Local Characteristics	
CPRATIO	Ratio of the county's Catholic residents to Protestant residents
REL	Proportion of the county population that are religious adherents
Education	Proportion of the county population over age 25 that has completed a bachelor's degree or higher
Married	Proportion of the county population over age 15 that is married
Income	Natural logarithm of the ZIP code-level median household income
Urban	Proportion of the county population that lives in urban area
Total population	Natural logarithm of the county total population
Over65	Proportion of the county population over age 65
Male-female ratio	Ratio of the county's male residents to female residents
Minority	Proportion of the county non-white residents
Crime rate	Crime rates of part I offenses excluding arson expressed as the number of crimes per 10,000 residents per year
Republic	Indicator that equals one if the Republic partisanship index is equal or greater than 50%
HPA	ZIP code house price appreciation in the two years prior to loan origination year. County level data used if ZIP code index is not available
Unemployment	Unemployment rate of the county
Loan characteristics	
Interest rate	Loan interest rate at origination
FICO	Natural logarithm of borrower's FICO credit score at loan origination
Balance	Natural logarithm of the initial loan balance
LTV	Loan-to-value ratio at loan origination
ARM	Indicator that equals one if the loan is an adjustable rate mortgage
Option ARM	Indicator that equals one if the loan has an ARM convertibility clause
Negative amortization	Indicator that equals one if the loan allows negative amortization
Low or no doc.	Indicator that equals one if the loan is originated with no or limited documentation
Prepayment penalty	Indicator that equals one if the loan would be assessed a penalty on any early voluntary prepayment
Cash-out	Indicator that equals one if the loan purpose is cash out refinancing
No-cash-out	Indicator that equals one if the loan purpose is no cash out refinancing
Investment	Indicator that equals one if the loan occupancy status is investment
Second-home	Indicator that equals one if the loan occupancy status is second-home
Default	Indicator that equals one if the loan becomes 90 days or more delinquent using MBA method in the first three years after origination

The table reports the variable definitions used in the empirical analysis part.

Table 2
Descriptive Statistics

	N	Mean	SD	P10	P25	P50	P75	P90
Simultaneous second liens (loan level)								
Misreported second (%)	3,031,921	7.15	25.76					
Reported second (%)	3,031,921	13.91	34.61					
Simul. second (%)	3,031,921	21.06	40.77					
Loan characteristics (loan level)								
Interest rate (%)	3,026,958	6.41	2.12	2.75	5.88	6.50	7.55	8.75
FICO	2,993,630	681.78	72.60	578.00	632.00	688.00	739.00	776.00
Balance (ln)	3,031,921	12.40	0.73	11.44	11.87	12.40	12.98	13.31
LTV (%)	3,031,921	74.82	13.34	56.65	70.00	80.00	80.00	90.00
ARM (%)	3,031,921	58.87	49.21					
Option ARM (%)	3,031,921	0.24	4.84					
Negative amortization (%)	3,031,921	13.58	34.26					
Low or no doc. (%)	3,031,921	65.42	47.56					
Prepayment penalty (%)	3,031,921	40.99	49.18					
Cash-out (%)	3,031,921	40.66	49.12					
No-cash-out (%)	3,031,921	14.73	35.44					
Investment (%)	3,031,921	10.27	30.36					
Second-home (%)	3,031,921	3.55	18.49					
Default (%)	3,014,278	23.51	42.40					
Local characteristics (county-year level)								
CPRATIO	8,716	0.51	0.88	0.01	0.04	0.19	0.54	1.29
REL (%)	8,716	59.59	17.64	36.62	45.90	58.51	72.42	89.21
Crime rate (per 10,000 residents)	8,228	279.34	150.46	113.14	161.38	244.84	358.32	496.85
Local characteristics (county-month level)								
Unemployment (%)	71,156	5.14	1.73	3.20	3.90	4.90	6.00	7.50
Local characteristics (county level)								
Education (%)	3,035	16.64	7.51	9.70	11.20	14.40	19.30	26.60
Married (%)	3,035	60.39	5.10	53.90	57.90	61.30	63.90	66.00
Urban (%)	3,035	40.62	30.54	0.00	12.93	40.32	64.65	84.67
Total population (ln)	3,035	10.49	1.11	9.52	9.52	10.15	11.05	12.09
Over65 (%)	3,035	14.71	4.05	9.90	12.10	14.40	17.00	20.00
Male-female ratio	3,035	0.98	0.06	0.92	0.94	0.97	1.00	1.05
Minority (%)	3,035	15.32	15.80	2.02	3.34	8.71	22.96	37.90
Republic	3,015	0.23	0.42					
Local characteristics (ZIP code-year level)								
HPA	76,535	18.45	14.28	4.23	7.55	13.93	26.56	39.99
Local characteristics (ZIP code level)								
Income (ln)	26,517	10.57	0.34	10.17	10.34	10.53	10.77	11.02

The table presents descriptive statistics for the variables used in our study, covering the sample period from 2005 to 2007. For all continuous variables, we report the number of observations, mean, standard deviation, 25th percentile, 50th percentile, and 75th percentile. For all dummy variables, we report the number of observations, mean, and standard deviation. The variables are winsorized at the 0.5 percent level. (ln) indicates that the value of the variable is the natural logarithm of the original value. (%) indicates that the value of the variable is expressed as a percentage.

Table 3
Local Socioeconomic Characteristics and Second-lien Misrepresentation

	(1)	(2)	(3)	(4)	(5)
CPRATIO	0.0017***				0.0014***
	(0.000)				(0.000)
REL	0.0001				-0.0001
	(0.000)				(0.000)
Crime rate	0.0008***				0.0008**
	(0.000)				(0.000)
Republic	0.0003				0.0009
	(0.001)				(0.001)
Education		0.0004			0.0005*
		(0.000)			(0.000)
Income		-0.0008***			-0.0004**
		(0.000)			(0.000)
Unemployment			-0.0002		-0.0008
			(0.000)		(0.001)
HPA			0.0017***		0.0017***
			(0.001)		(0.001)
Urban			-0.0003		-0.0007**
			(0.000)		(0.000)
Total population			0.0014***		0.0010***
			(0.000)		(0.000)
Married				-0.0001	0.0002
				(0.000)	(0.000)
Over65				0.0002	0.0006**
				(0.000)	(0.000)
Male-female ratio				-0.0003	-0.0000
				(0.000)	(0.000)
Minority				0.0010**	0.0006
				(0.000)	(0.000)
Simul. Second	0.3125***	0.3130***	0.3126***	0.3126***	0.3130***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Interest rate	-0.0072***	-0.0074***	-0.0073***	-0.0073***	-0.0072***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
FICO	-0.0063***	-0.0063***	-0.0062***	-0.0063***	-0.0062***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Balance	0.0049***	0.0051***	0.0048***	0.0050***	0.0048***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
LTV	0.0056***	0.0055***	0.0056***	0.0056***	0.0056***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)

Table 3 (continued)

	(1)	(2)	(3)	(4)	(5)
ARM	0.0087***	0.0088***	0.0086***	0.0086***	0.0088***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Negative amortization	-0.0037***	-0.0044***	-0.0038***	-0.0040***	-0.0041***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Option ARM	-0.0227***	-0.0227***	-0.0227***	-0.0227***	-0.0227***
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
Prepayment penalty	-0.0079***	-0.0080***	-0.0079***	-0.0078***	-0.0082***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Low or no doc.	-0.0078***	-0.0079***	-0.0080***	-0.0078***	-0.0080***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Cash-out	0.0184***	0.0181***	0.0184***	0.0184***	0.0182***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
No-cash-out	0.0156***	0.0156***	0.0158***	0.0157***	0.0158***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Investment	0.0259***	0.0260***	0.0259***	0.0259***	0.0258***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Second-home	0.0157***	0.0158***	0.0157***	0.0157***	0.0158***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Originator FE	Y	Y	Y	Y	Y
State FE	Y	Y	Y	Y	Y
Half-year FE	Y	Y	Y	Y	Y
Observations	2,453,157	2,389,104	2,481,611	2,486,167	2,353,016
Adj. R^2	0.317	0.318	0.317	0.317	0.318

The table shows OLS estimates from regressions where the dependent variable takes a value of one if the loan is recognized as second-lien misreported and zero otherwise. All continuous variables are winsorized at the 0.5 percent level and then standardized. Standard errors clustered at the county level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 4
Gambling Preference and Second-lien Misrepresentation - Borrower Motives

	(1) Primary	(2) Non-primary	(3) Purchase	(4) Refinance
CPRATIO	0.0014***	0.0008	0.0022***	0.0006*
	(0.000)	(0.001)	(0.001)	(0.000)
REL	-0.0001	-0.0003	-0.0005	0.0001
	(0.000)	(0.001)	(0.001)	(0.000)
Crime rate	0.0010**	0.0009	0.0016**	0.0003
	(0.000)	(0.001)	(0.001)	(0.000)
Republic	0.0009	0.0005	0.0012	0.0005
	(0.001)	(0.001)	(0.001)	(0.000)
Education	0.0007**	-0.0011*	0.0008	0.0003
	(0.000)	(0.001)	(0.001)	(0.000)
Income	-0.0011***	0.0016***	-0.0003	-0.0003**
	(0.000)	(0.000)	(0.000)	(0.000)
Unemployment	-0.0008	-0.0011*	-0.0010	-0.0001
	(0.001)	(0.001)	(0.001)	(0.000)
HPA	0.0018***	-0.0000	0.0021**	0.0019***
	(0.001)	(0.001)	(0.001)	(0.000)
Urban	-0.0008**	0.0010	-0.0008	-0.0006**
	(0.000)	(0.001)	(0.001)	(0.000)
Total population	0.0011***	-0.0010	0.0020***	0.0007**
	(0.000)	(0.001)	(0.001)	(0.000)
Married	0.0004	-0.0004	0.0008	-0.0002
	(0.000)	(0.001)	(0.001)	(0.000)
Over65	0.0006**	0.0004	0.0001	0.0005**
	(0.000)	(0.000)	(0.000)	(0.000)
Male-female ratio	-0.0001	0.0007	-0.0001	0.0003
	(0.000)	(0.001)	(0.000)	(0.000)
Minority	0.0007	-0.0006	0.0009	0.0003
	(0.000)	(0.001)	(0.001)	(0.000)
Loan chars controls	Y	Y	Y	Y
Originator FE	Y	Y	Y	Y
State FE	Y	Y	Y	Y
Half-year FE	Y	Y	Y	Y
Observations	2,052,270	300,095	1,007,280	1,344,733
Adj. R^2	0.312	0.418	0.315	0.382

The table shows OLS estimates from regressions where the dependent variable takes a value of one if the loan is recognized as second-lien misreported and zero otherwise. We look at subsamples by borrower motives: primary or non-primary (column (1) and (2)) and purchase or refinance (column (3) and (4)). All continuous variables are winsorized at the 0.5 percent level and then standardized. Standard errors clustered at the county level are reported in parentheses. *$p < 0.10$, **$p < 0.05$, and ***$p < 0.01$.

Table 5
Gambling Preference and Second-Lien Misrepresentation – Credit Expansion

	(1) High Income	(2) Low Income	(3) High FICO	(4) Low FICO
CPRATIO	0.0013**	0.0017***	0.0012**	0.0017***
	(0.001)	(0.001)	(0.001)	(0.000)
REL	0.0001	-0.0010**	-0.0008*	0.0003
	(0.000)	(0.000)	(0.000)	(0.000)
Crime rate	0.0008	0.0012**	0.0012**	0.0005
	(0.000)	(0.001)	(0.001)	(0.000)
Republic	0.0013*	0.0002	0.0007	0.0011
	(0.001)	(0.001)	(0.001)	(0.001)
Education	0.0007**	-0.0002	0.0003	0.0009**
	(0.000)	(0.001)	(0.000)	(0.000)
Income	0.0001	-0.0017**	-0.0001	-0.0008**
	(0.000)	(0.001)	(0.000)	(0.000)
Unemployment	-0.0006	-0.0012**	-0.0013**	-0.0002
	(0.001)	(0.001)	(0.001)	(0.001)
HPA	0.0025***	-0.0004	0.0021***	0.0022***
	(0.001)	(0.001)	(0.001)	(0.001)
Urban	-0.0004	-0.0012**	-0.0003	-0.0011***
	(0.000)	(0.000)	(0.000)	(0.000)
Total population	0.0007	0.0019***	0.0016***	0.0004
	(0.000)	(0.001)	(0.001)	(0.001)
Married	0.0000	0.0007	0.0007	-0.0007*
	(0.000)	(0.001)	(0.001)	(0.000)
Over65	0.0007**	0.0002	0.0003	0.0005
	(0.000)	(0.000)	(0.000)	(0.000)
Male-female ratio	0.0003	-0.0006*	0.0005	-0.0003
	(0.000)	(0.000)	(0.000)	(0.000)
Minority	0.0009*	-0.0000	0.0007	0.0003
	(0.001)	(0.001)	(0.001)	(0.000)
Loan chars controls	Y	Y	Y	Y
Originator FE	Y	Y	Y	Y
State FE	Y	Y	Y	Y
Half-year FE	Y	Y	Y	Y
Observations	1,809,909	542,366	1,308,243	1,044,300
Adj. R^2	0.315	0.329	0.298	0.358

The table shows OLS estimates from regressions where the dependent variable takes a value of one if the loan is recognized as second-lien misreported and zero otherwise. We look at subsamples related to credit expansion argument: high or low income ((1) and (2)) and high or low credit score ((3) and (4)). The high or low income subsamples are defined by comparing local household median income with the sample median at the ZIP code level. The high or low credit score subsamples are divided by a FICO score of 670. All continuous variables are winsorized at the 0.5 percent level and then standardized. Standard errors clustered at the county level are reported in parentheses. $^*p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 6
Gambling Preference and Second-Lien Misrepresentation – Housing Expectation

	(1) High HPA	(2) Low HPA
CPRATIO	0.0018***	0.0008
	(0.000)	(0.001)
REL	-0.0010***	0.0006
	(0.000)	(0.000)
Crime rate	0.0002	0.0013**
	(0.000)	(0.001)
Republic	0.0027***	-0.0016*
	(0.001)	(0.001)
Education	0.0008*	0.0007*
	(0.000)	(0.000)
Income	-0.0007**	-0.0000
	(0.000)	(0.000)
Unemployment	-0.0004	-0.0004
	(0.001)	(0.001)
HPA	0.0005	0.0055***
	(0.001)	(0.001)
Urban	-0.0011**	-0.0004
	(0.000)	(0.000)
Total population	0.0017***	0.0008
	(0.000)	(0.001)
Married	0.0002	0.0004
	(0.000)	(0.001)
Over65	0.0003	0.0007*
	(0.000)	(0.000)
Male-female ratio	-0.0004	0.0006
	(0.000)	(0.000)
Minority	0.0008	0.0004
	(0.001)	(0.001)
Loan chars controls	Y	Y
Originator FE	Y	Y
State FE	Y	Y
Half-year FE	Y	Y
Observations	1,352,931	999,126
Adj. R^2	0.321	0.315

The table shows OLS estimates from regressions where the dependent variable takes a value of one if the loan is recognized as second-lien misreported and zero otherwise. We look at subsamples related to housing expectation: high or low HPA areas ((1) and (2)). The high and low HPA subsamples are defined based on the state-year median HPA, calculated from all available ZIP code-level data. All continuous variables are winsorized at the 0.5 percent level and then standardized. Standard errors clustered at the county level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 7
Discontinuities for Low-documentation Loans Around the Credit Threshold in All Areas

	(1) All loan	(2) Reported second	(3) Misreported second
Panel A. Jump Ratio in Number of Loans			
Est. 620+/620-	2.266	4.939	3.838
Panel B. Default Rate			
FICO$\geq$620 (β)	0.038	0.045	0.046
t-stat	(3.89)	(2.18)	(-0.84)

The table reports the discontinuities for low-documentation loans around a FICO score of 620 in all areas. Panel A presents the estimates from regressions where the dependent variable is the number of loans at each FICO score. Using local linear regressions of the RDD approach, we estimate the number of loans at FICO 620^{-} and FICO 620^{+} and compute the ratio of the estimated number of loans at FICO 620^{+} to 620^{-}. Panel B presents the estimates from regressions where the dependent variable takes a value of one if the loan becomes 90 days or more delinquent using the MBA method in the first three years after origination and zero otherwise. Using local linear regressions of the RDD approach with control variables, we estimate the difference in default rates between FICO 620^{-} and FICO 620^{+}. We perform the estimation for all loans, loans with correctly reported simultaneous seconds, and loans with misreported simultaneous seconds, and the results are reported in columns (1) to (3), respectively. Bias-corrected t-values (standard errors clustered at the county level) following Calonico et al. (2014) are reported in parentheses.

Table 8
Discontinuities for Low-documentation Loans Around the Credit Threshold in High and Low Gambling Preference Areas

	(1) High	(2) Low	(3) High-Low
Panel A. Jump Ratio in Number of Loans (Est. 620+/620-)			
All loan	2.240	2.289	-0.049
			(-0.52)
Reported second	5.498	4.581	0.917
			(1.23)
Misreported second	2.763	4.826	-2.063
			(-2.83)
Panel B. Default Rate			
All loan	0.039	0.037	0.001
	(3.45)	(2.29)	(0.11)
Reported second	0.012	0.064	-0.052
	(0.22)	(2.64)	(-1.36)
Misreported second	0.017	0.077	-0.061
	(0.12)	(-1.09)	(-1.31)

The table reports the discontinuities for low-documentation loans around a FICO score of 620 separately in high and low gambling preference areas. Panel A presents the estimates from regressions where the dependent variable is the number of loans at each FICO score. Using local linear regressions of the RDD approach, we estimate the number of loans at FICO 620^- and FICO 620^+ and compute the ratio of the estimated number of loans at FICO 620^+ to 620^-. Panel B presents the estimates from regressions where the dependent variable is the default dummy variable. Using local linear regressions of the RDD approach with control variables, we estimate the difference in default rates between FICO 620^- and FICO 620^+. Using the diff-in-disc approach, we estimate the difference between high and low gambling preference areas. We perform the estimation for all loans, loans with correctly reported simultaneous seconds, and loans with misreported simultaneous seconds. The results for loans in high, low, and the difference between high and low gambling preference areas are reported in columns (1) to (3), respectively. For discontinuities, bias-corrected t-values (standard errors clustered at the county level) following Calonico et al. (2014) are reported in parentheses. For the difference between high and low gambling preference areas, t-statistics based on heteroskedasticity-consistent standard errors are reported in parentheses.

Table 9
Outcomes of Misrepresentation Across Gambling Preference Levels

	(1) High CPRATIO	(2) Low CPRATIO
Misreported second	0.141***	0.100***
	(0.004)	(0.005)
Reported second	0.181***	0.111***
	(0.008)	(0.006)
REL	-0.006*	-0.001
	(0.004)	(0.002)
Crime rate	0.019***	0.001
	(0.005)	(0.003)
Republic	-0.014**	-0.007
	(0.007)	(0.006)
Education	0.003	0.001
	(0.004)	(0.003)
Income	-0.012***	-0.013***
	(0.003)	(0.002)
Unemployment	-0.004	0.009***
	(0.004)	(0.002)
HPA	0.042***	0.064***
	(0.006)	(0.007)
Urban	0.010**	-0.007***
	(0.004)	(0.002)
Total population	-0.014**	0.027***
	(0.006)	(0.004)
Married	0.022***	0.017***
	(0.003)	(0.003)
Over65	0.009*	-0.000
	(0.006)	(0.003)
Male-female ratio	0.001	-0.002
	(0.005)	(0.002)
Minority	0.007	0.006
	(0.005)	(0.004)
Interest rate	0.027***	0.037***
	(0.002)	(0.002)
FICO	-0.089***	-0.098***
	(0.002)	(0.002)
Balance	0.022***	0.024***
	(0.003)	(0.001)
LTV	0.056***	0.047***
	(0.002)	(0.003)

Table 10 (continued)

	(1) High CPRATIO	(2) Low CPRATIO
ARM	0.047***	0.033***
	(0.003)	(0.002)
Negative amortization	0.023***	0.049***
	(0.004)	(0.006)
Option ARM	0.051***	0.037***
	(0.011)	(0.009)
Prepayment penalty	0.065***	0.048***
	(0.003)	(0.003)
Low or no doc.	0.055***	0.052***
	(0.002)	(0.003)
Cash-out	-0.019***	-0.030***
	(0.003)	(0.002)
No-cash-out	0.025***	-0.006*
	(0.005)	(0.003)
Investment	0.023***	0.063***
	(0.005)	(0.004)
Second-home	0.004	0.011**
	(0.005)	(0.005)
Originator FE	Y	Y
State FE	Y	Y
Half-year FE	Y	Y
Observations	1,149,680	1,201,389
Adj. R^2	0.252	0.207

The table shows OLS estimates from regressions where the dependent variable takes a value of one if the loan becomes 90 days or more delinquent using the MBA method in the first three years after origination and zero otherwise. We look at subsamples by gambling preference: high or low CPRATIO (column (1) and (2)). All continuous variables are winsorized at the 0.5 percent level and then standardized. All specifications include state, half-year, and originator fixed effects. Standard errors clustered at the county level are reported in parentheses. *$p < 0.10$, **$p < 0.05$, and ***$p < 0.01$.

Table 10
Local Socioeconomic Characteristics and Mortgage Misrepresentation - Probit Model in Simultaneous Seconds Sample

	(1)	(2)	(3)	(4)	(5)
CPRATIO	0.007***				0.005***
	(0.002)				(0.002)
REL	0.000				-0.001
	(0.001)				(0.001)
Crime rate	0.003***				0.002
	(0.001)				(0.002)
Republic	0.000				0.003
	(0.002)				(0.002)
Education		0.001			0.003**
		(0.001)			(0.001)
Income		-0.003***			-0.002*
		(0.001)			(0.001)
Unemployment			0.002		0.001
			(0.001)		(0.002)
HPA			0.004**		0.004**
			(0.002)		(0.002)
Urban			-0.001		-0.003**
			(0.001)		(0.001)
Total population			0.007***		0.005***
			(0.001)		(0.002)
Married				0.000	0.001
				(0.001)	(0.001)
Over65				-0.000	0.001
				(0.001)	(0.001)
Male-female ratio				-0.001	-0.000
				(0.001)	(0.001)
Minority				0.005***	0.003*
				(0.001)	(0.002)
Loan chars controls	Y	Y	Y	Y	Y
Originator FE	Y	Y	Y	Y	Y
State FE	Y	Y	Y	Y	Y
Half-year FE	Y	Y	Y	Y	Y
Observations	462,669	448,139	467,972	468,625	441,794

The table shows OLS estimates from regressions where the dependent variable takes a value of one if the loan is recognized as second-lien misreported and zero otherwise. The sample is restricted to the loans with simultaneous second liens. All continuous variables, except for the crime rates, are winsorized at the 0.5 percent level and then standardized. Standard errors clustered at the county level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 11
Local Socioeconomic Characteristics and Mortgage Misrepresentation - Probit Model in Whole Sample

	(1)	(2)	(3)	(4)	(5)
CPRATIO	0.0017**				0.0014**
	(0.001)				(0.001)
REL	-0.0004				-0.0005
	(0.000)				(0.000)
Crime rate	0.0001				0.0018***
	(0.000)				(0.001)
Republic	0.0007				-0.0007
	(0.001)				(0.001)
Education		0.0003			0.0001
		(0.001)			(0.001)
Income		-0.0003			-0.0003
		(0.000)			(0.000)
Unemployment			-0.0020***		-0.0026***
			(0.000)		(0.001)
HPA			0.0041***		0.0040***
			(0.001)		(0.001)
Urban			0.0000		-0.0004
			(0.000)		(0.000)
Total population			0.0003		0.0006
			(0.000)		(0.000)
Married				0.0012*	0.0022***
				(0.001)	(0.001)
Over65				-0.0008**	0.0000
				(0.000)	(0.000)
Male-female ratio				-0.0005	0.0000
				(0.000)	(0.000)
Minority				0.0006	0.0005
				(0.001)	(0.001)
Prevalence of simul.	0.1751***	0.1780***	0.1761***	0.1723***	0.1695***
second	(0.005)	(0.006)	(0.006)	(0.006)	(0.006)
Loan chars controls	Y	Y	Y	Y	Y
Originator FE	Y	Y	Y	Y	Y
State FE	Y	Y	Y	Y	Y
Half-year FE	Y	Y	Y	Y	Y
Observations	2,342,532	2,280,584	2,368,874	2,372,952	2,247,430

The table shows OLS estimates from regressions where the dependent variable takes a value of one if the loan is recognized as second-lien misreported and zero otherwise. All continuous variables, except for the crime rates, are winsorized at the 0.5 percent level and then standardized. Standard errors clustered at the county level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 12
Gambling Preference and Mortgage Misrepresentation - Causal Forest

	(1) Misreported second
CPRATIO	0.0016**
	(0.0007)

The table shows causal forest estimates from regressions where the dependent variable takes a value of one if the loan is recognized as second-lien misreported and zero otherwise. All control variables in Table 3 are used for growing trees and forests. Before growing the trees, all continuous variables are winsorized at the 0.5 percent level and then standardized. A half-year variable is created, starting from the first half of 2005 as 1, and used as a variable in growing trees and forests. Observations are clustered by state, and each unit is given the same weight (so that larger clusters receive more weight). 2000 trees are grown in the causal forest. The fraction used for determining splits (honesty) is 50 percent. Standard errors are reported in parentheses. *$p < 0.10$, **$p < 0.05$, and ***$p < 0.01$.

Table 13
Alternative Gambling Preference Measures

	(1)	(2)
Gambling Industry Employment	0.0031***	
	(0.001)	
CPRATIO (adjusted)		0.0011**
		(0.000)
REL	0.0025*	0.0001
	(0.001)	(0.000)
Crime rate	0.0039**	0.0008*
	(0.002)	(0.000)
Republic	-0.0016	0.0008
	(0.002)	(0.001)
Education	-0.0004	0.0004
	(0.002)	(0.000)
Income	0.0005	-0.0004*
	(0.002)	(0.000)
Unemployment	-0.0011	-0.0008
	(0.001)	(0.001)
HPA	0.0016	0.0017***
	(0.002)	(0.001)
Urban	0.0005	-0.0004
	(0.002)	(0.000)
Total population	0.0009	0.0009**
	(0.002)	(0.000)
Married	0.0003	0.0002
	(0.001)	(0.000)
Over65	-0.0000	0.0005*
	(0.001)	(0.000)
Male-female ratio	0.0020	-0.0000
	(0.001)	(0.000)
Minority	-0.0026*	0.0006
	(0.001)	(0.000)
Loan chars controls	Y	Y
Originator FE	Y	Y
State FE	Y	Y
Half-year FE	Y	Y
Observations	1,032,077	2,308,542
Adj. R^2	0.318	0.318

The table shows OLS estimates from regressions where the dependent variable takes a value of one if the loan is recognized as second-lien misreported and zero otherwise. The local controls in column (1) are aggregated to CBSA level. The local controls in column (2) follow the baseline specification and remain at the same geographical level as in the main regressions. All continuous variables are winsorized at the 0.5 percent level and then standardized. Standard errors are clustered at the CBSA level for column (1) and at the county level for column (2). Standard errors are reported in parentheses. $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

Table 14
Local Characteristics and Loan Collateral Fundamentals around Discontinuity in Low-Documentation Loans

	FICO≥620 (β))	t-stat
CPRATIO	-0.041	(-0.035)
REL	0.007	(0.741)
Crime rate	-5.528	(-0.416)
Republic	0.011	(0.049)
Education (%)	0.619	(1.561)
Income	0.024	(1.460)
Unemployment (%)	0.024	(-0.210)
HPA	-0.002	(-0.102)
Urban (%)	-0.278	(0.330)
Total population (ln)	-0.045	(0.038)
Married (%)	0.191	(0.135)
Over65 (%)	-0.386	(-0.883)
Male-female ratio	0.001	(0.147)
Minority (%)	-0.032	(0.123)
Balance	0.045	(0.868)
LTV (%)	0.637	(-0.329)

The table reports estimates from regressions of local characteristics and loan collateral fundamentals around a FICO score of 620 for low-documentation loans. Local linear regressions using the RDD approach are employed to estimate the differences in these predetermined variables between FICO 620^{-} and FICO 620^{+}. The bias-corrected t-values (standard errors clustered at the county level) following Calonico et al. (2014) are reported in parentheses.

Table 15
Discontinuities in Reported and Misreported Second-Lien Rates around the Credit Threshold in Low-Documentation Loans

		(1) All	(2) High	(3) Low	(4) High-Low
Panel A. Misreported second					
Raw	FICO≥620 (β)	0.042	0.014	0.065	-0.051
	t-stat	(6.49)	(0.69)	(9.13)	-(5.19)
Control	FICO≥620 (β)	0.032	0.009	0.052	-0.043
	t-stat	(5.52)	(0.26)	(7.86)	-(4.49)
Control+FEs	FICO≥620 (β)	0.010	0.006	0.015	-0.008
	t-stat	(2.97)	(0.97)	(3.33)	-(1.14)
Panel B. Reported second					
Raw	FICO≥620 (β)	0.090	0.094	0.087	0.006
	t-stat	(11.22)	(7.38)	(8.58)	(0.57)
Control	FICO≥620 (β)	0.078	0.087	0.070	0.017
	t-stat	(11.13)	(7.32)	(8.84)	(1.85)
Control+FEs	FICO≥620 (β)	0.062	0.066	0.058	0.008
	t-stat	(8.35)	(5.84)	(6.23)	(0.89)

The table reports discontinuities in reported and misreported second-lien rates around the FICO score threshold of 620 for low-documentation loans. Panel A presents the estimates for misreported second-lien rates, where the dependent variable is the dummy for misreported seconds, and Panel B presents the estimates for correctly reported second-lien rates, where the dependent variable is the dummy for correctly reported seconds. Local linear regressions from the RDD approach are used to estimate the difference in rates between FICO 620^{-} and FICO 620^{+}, . Using the diff-in-disc approach, we further estimate the difference between high and low gambling preference areas. Estimations are conducted without controls, with controls, and with both controls and the fixed effects from the baseline regressions. For discontinuities, bias-corrected t-values (standard errors clustered at the county level) following Calonico et al. (2014) are reported in parentheses. For the difference between high and low gambling preference areas, t-statistics based on heteroskedasticity-consistent standard errors are reported in parentheses.

Table 16
Alternative Default Measures for Loan Performance Tests

	(1)	(2)	(3)	(4)	(5)	(6)
	90+ delinq in 2 year		60+ delinq in 3 year		B/F/REO in 3 year	
	High	Low	High	Low	High	Low
Misreported second	0.076***	0.055***	0.143***	0.102***	0.124***	0.092***
	(0.003)	(0.004)	(0.003)	(0.005)	(0.004)	(0.005)
Reported second	0.105***	0.058***	0.183***	0.114***	0.145***	0.090***
	(0.006)	(0.004)	(0.007)	(0.006)	(0.007)	(0.006)
REL	-0.004*	-0.001	-0.007*	0.001	-0.004	-0.001
	(0.002)	(0.001)	(0.004)	(0.002)	(0.003)	(0.002)
Crime rate	0.010***	0.003	0.018***	0.000	0.019***	0.003
	(0.003)	(0.002)	(0.005)	(0.003)	(0.004)	(0.002)
Republic	-0.004	-0.003	-0.017**	-0.007	-0.013**	-0.004
	(0.005)	(0.004)	(0.007)	(0.006)	(0.006)	(0.005)
Education	0.002	0.002	0.002	-0.000	0.003	0.002
	(0.003)	(0.002)	(0.004)	(0.003)	(0.003)	(0.003)
Income	-0.008***	-0.010***	-0.013***	-0.014***	-0.009***	-0.010***
	(0.002)	(0.001)	(0.003)	(0.002)	(0.002)	(0.002)
Unemployment	0.000	0.010***	-0.005	0.008***	-0.003	0.009***
	(0.003)	(0.002)	(0.004)	(0.002)	(0.004)	(0.002)
HPA	0.022***	0.037***	0.041***	0.065***	0.033***	0.051***
	(0.005)	(0.005)	(0.006)	(0.007)	(0.005)	(0.005)
Urban	0.007**	-0.004***	0.011***	-0.009***	0.006	-0.004**
	(0.003)	(0.002)	(0.004)	(0.002)	(0.004)	(0.002)
Total population	-0.006*	0.016***	-0.013**	0.027***	-0.012**	0.018***
	(0.003)	(0.003)	(0.006)	(0.004)	(0.005)	(0.004)
Married	0.011***	0.011***	0.022***	0.017***	0.018***	0.014***
	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)	(0.003)
Over65	0.007**	0.001	0.010*	-0.001	0.007	0.003
	(0.004)	(0.002)	(0.006)	(0.003)	(0.005)	(0.003)
Male-female ratio	-0.001	-0.001	0.001	-0.002	-0.000	-0.002
	(0.003)	(0.002)	(0.005)	(0.002)	(0.004)	(0.002)
Minority	0.003	0.003	0.006	0.007	0.003	0.002
	(0.003)	(0.003)	(0.005)	(0.004)	(0.005)	(0.004)
Loan chars controls	Y	Y	Y	Y	Y	Y
Originator FE	Y	Y	Y	Y	Y	Y
State FE	Y	Y	Y	Y	Y	Y
Half-year FE	Y	Y	Y	Y	Y	Y
Observations	1,149,680	1,201,389	1,149,680	1,201,389	1,149,680	1,201,389
Adj. R^2	0.171	0.148	0.260	0.220	0.176	0.143

The table shows OLS estimates from regressions in which the dependent variable uses different measures of default. We look at subsamples by gambling preference: high or low CPRATIO. Column (1) and (2) uses 90 days or more delinquency using MBA method in the first two years after origination. Column (3) and (4) uses 60 days or more delinquency using MBA method in the first two years after origination. Column (5) and (6) uses bankruptcy/foreclosure/REO in the first three years after origination. All continuous variables are winsorized at 0.5 percent level and then standardized. Standard errors clustered at the county level are reported in the parentheses. $^*p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

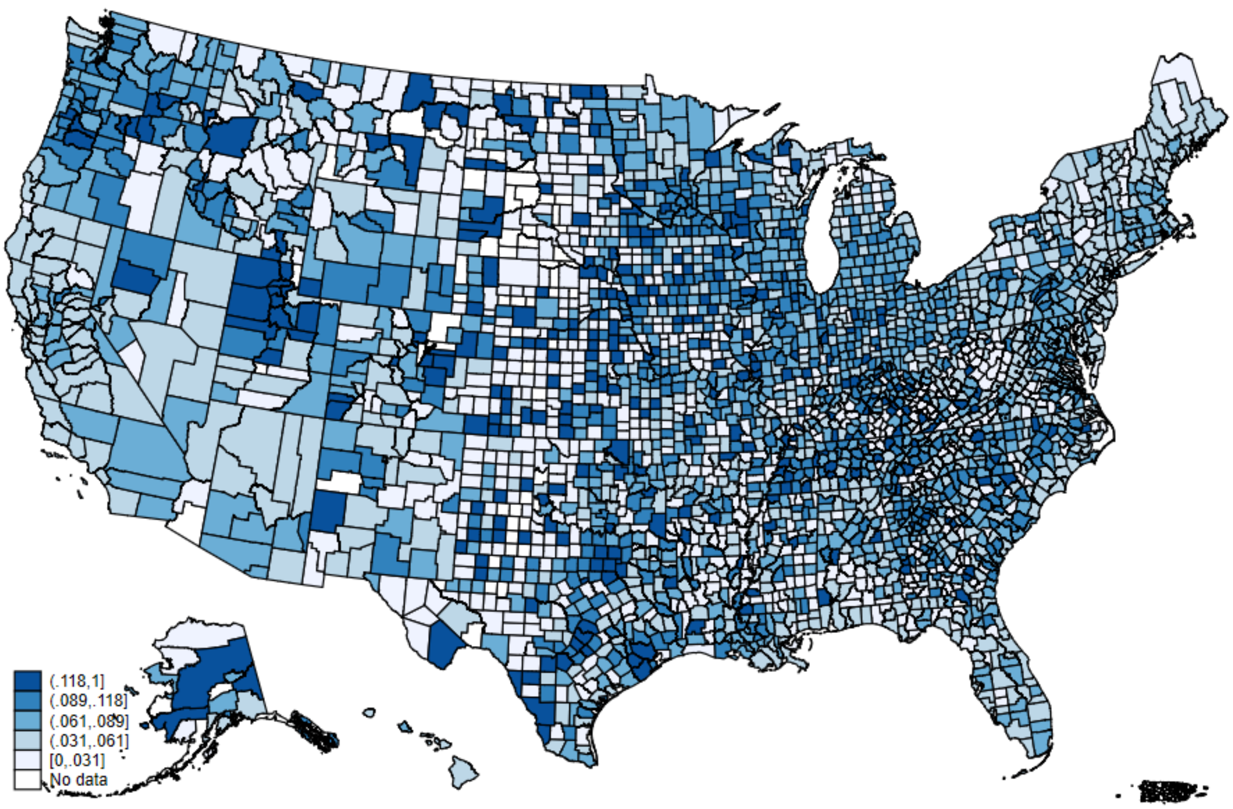

Figure 1
Second-lien Misrepresentations Distribution
The figure plots the county level proportion of second-lien misrepresentation in all loans.

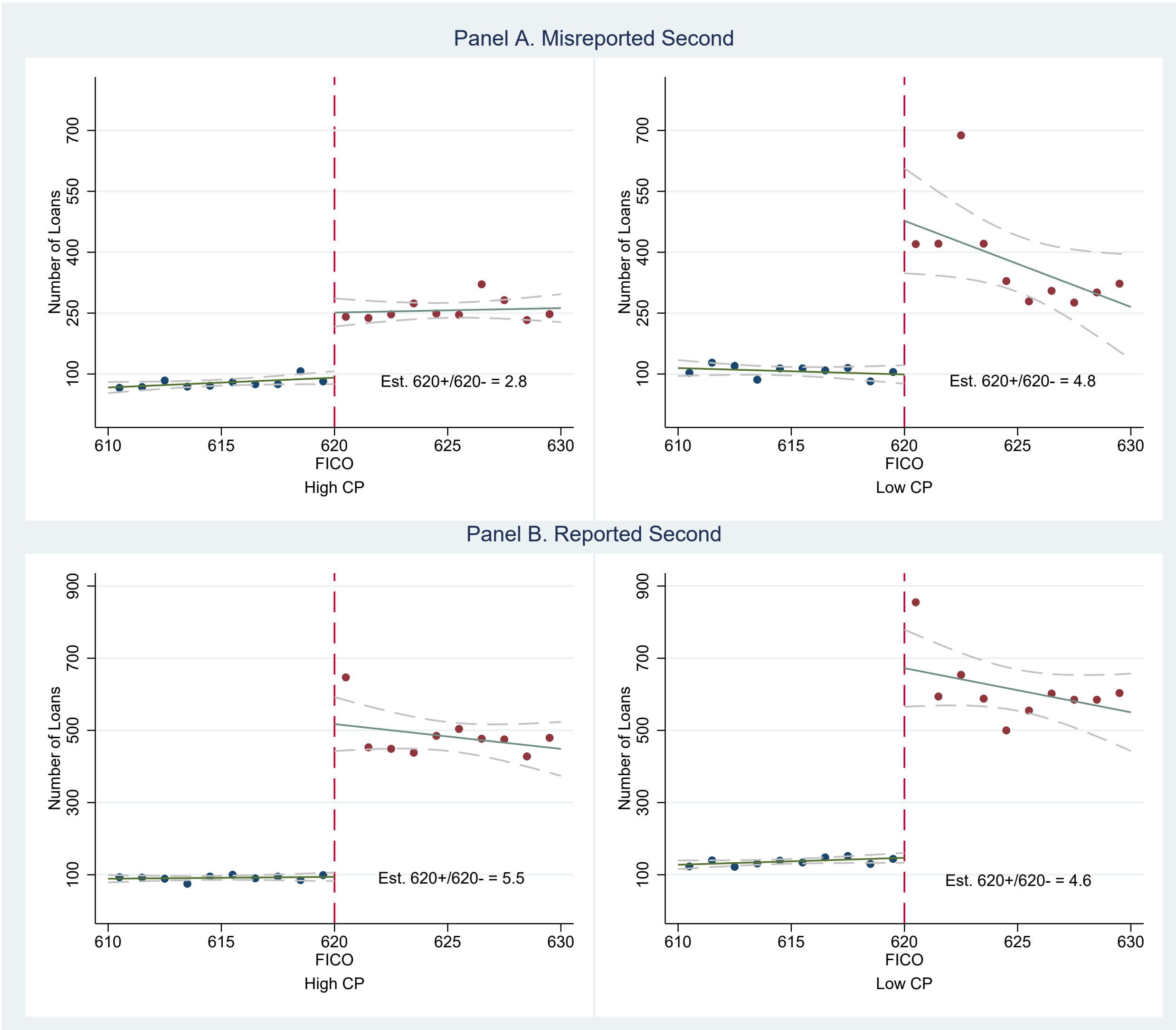

Figure 2
Discontinuities in the Number of Low-Documentation Loans Around Credit Score Thresholds by Gambling Preference
The figure plots discontinuities in the number of low-documentation loans around the FICO score threshold of 620 in high and low gambling preference areas. Panel A reports the loans with misreported seconds, and Panel B reports loans with correctly reported seconds. The circles represent the average number of loans for each credit score between 610 and 629. The solid lines show the linear fit on either side of the credit score threshold, with dashed lines denoting the 95% confidence interval. The left subfigures report results for loans in high gambling preference areas, while the right subfigures present results for loans in low gambling preference areas. As shown, the jump ratio for loans with misreported seconds is much smaller in high gambling preference areas than in low gambling preference areas, while the jump ratio for loans with correctly reported seconds is similar across the two sets of areas.

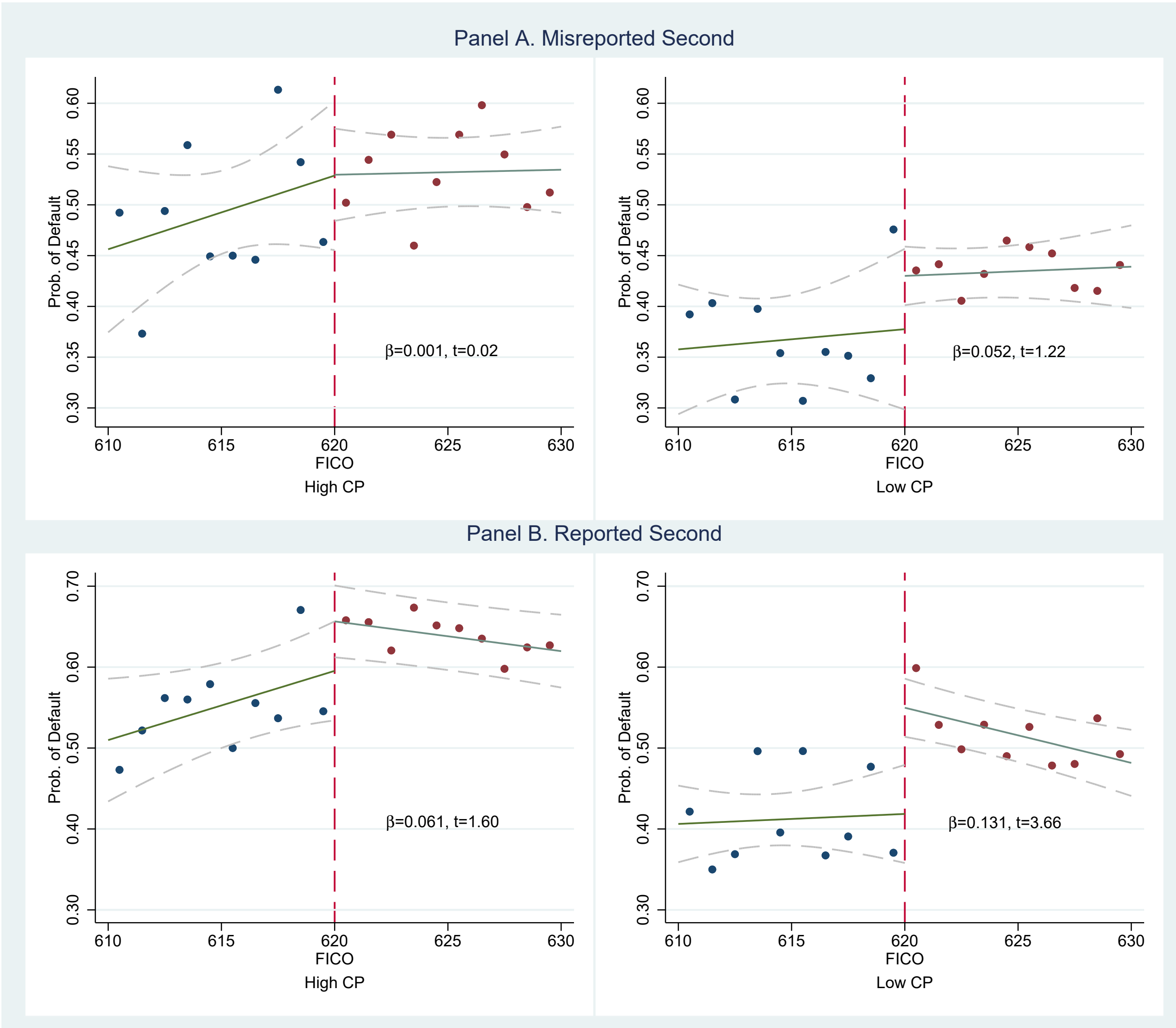

Figure 3
Discontinuities in Default Rates for Low-Documentation Loans Around Credit Score Thresholds by Gambling Preference

The figure plots discontinuities in default rates for low-documentation loans around the FICO score threshold of 620 in high and low gambling preference areas. Panel A reports the loans with misreported seconds, and Panel B reports loans with correctly reported seconds. The circles represent the average number of loans for each credit score between 610 and 629. The solid lines show the linear fit on either side of the credit score threshold, with dashed lines denoting the 95% confidence interval. The left subfigures report results for loans in high gambling preference areas, while the right subfigures present results for loans in low gambling preference areas. As shown, for loans with misreported seconds, the jump in default rates is insignificant in both groups, although it is larger in magnitude in low gambling preference areas. For loans with correctly reported seconds, the jump in default rates is also greater in low gambling preference areas.

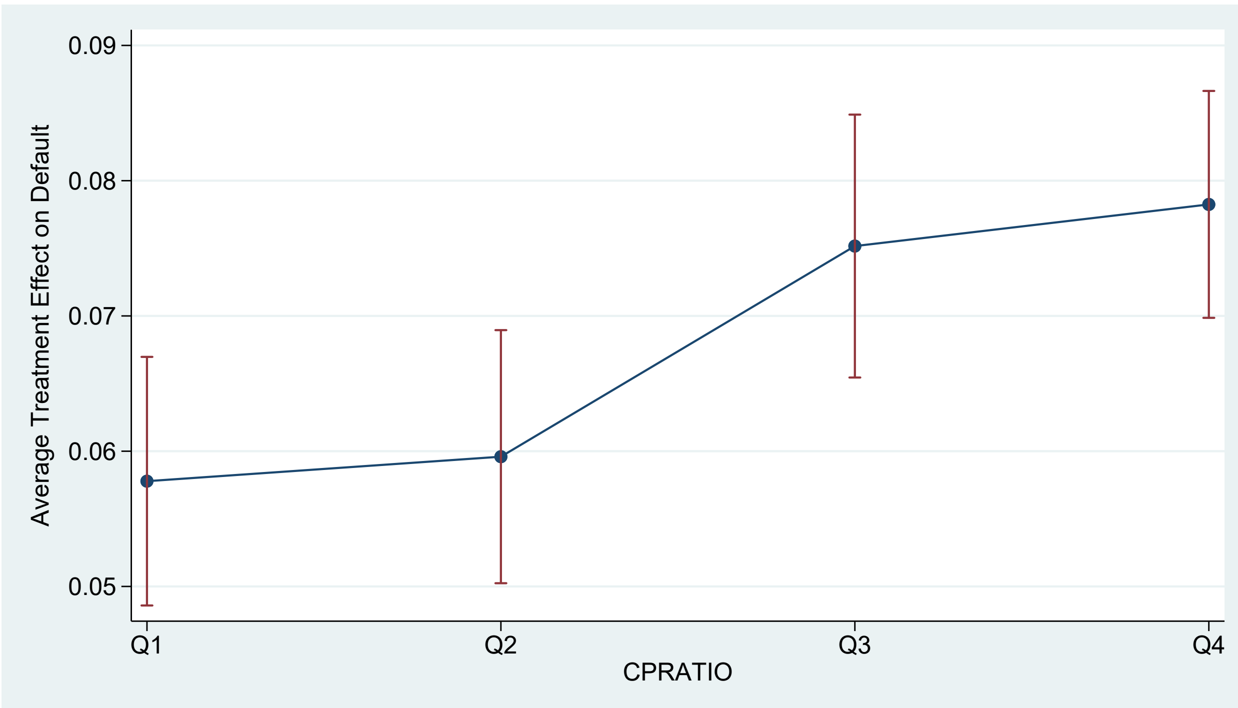

Figure 4
Heterogeneous Treatment Effects of Second-lien Misrepresentation on Default Conditioned on Different Levels of Gambling Preference
The figure plots the heterogeneous treatment effects of second-lien misrepresentation on default, conditioned on different levels of gambling preference, using the causal forest approach. The dependent variable is an indicator that takes a value of one if the loan becomes 90 days or more delinquent using the MBA method in the first three years after origination. The average treatment effect of mortgage misrepresentation is estimated at different levels of local gambling preference (four quarters divided by CPRATIO). All control variables in column (3) of Table 9 are used for growing trees and forests. Before growing the trees, all continuous variables are winsorized at the 0.5 percent level and then standardized. A half-year variable is created, starting from the first half of 2005 as 1, and used as a variable in growing trees and forests. Observations are clustered by state, and each unit is given the same weight (so that larger clusters receive more weight). 2000 trees are grown in the causal forest. The fraction used for determining splits (honesty) is 50 percent.

Appendix

A Alternative Gambling Preference Measures

In this section, we provide detailed information regarding the use of gambling industry employment as alternative gambling preference measure.

A.1 Data and Measure Construction

Since *CPRATIO* may capture broader religious influences, such as attitudes toward forgiveness and dishonesty, we also employ more direct measures of local gambling preference. Specifically, we use per capita employment in the gambling industry to quantify local gambling engagement.

To construct the measure, we extract employment data at the core-based statistical area (CBSA) level[1] for the years 2005 through 2007 from the County Business Patterns (CBP) dataset provided by the U.S. Census Bureau. To identify gambling-specific industries, we retain observations with NAICS code 7132, which includes 71321 for Casinos (except Casino Hotels) and 71329 for Other Gambling Industries. These counts are then normalized by dividing by the total population of the area. Population figures for 2005 to 2007 are linearly interpolated based on CBSA-level data from the 2000 and 2010 Census.

In addition to the gambling preference measure described above, we construct supplementary proxies. Because CBP classifies businesses based on their primary services, casino hotels (NAICS code 72110) are excluded from the gambling industry category. However, as these businesses also provide gambling services to the public, we extract employment data for casino hotels separately, normalizing it by total population as well.

Since the CBP dataset includes only employer-based businesses, some small gambling venues without formal employees may be omitted. To account for these, we incorporate data from the Nonemployer Statistics (NES) datasets, also maintained by the U.S. Census Bureau. We retain NES observations with NAICS code 7132[2]. The number of nonemployer gambling establishments is likewise normalized by CBSA population.

While the CBP dataset records the number of establishments without restrictions, the employment data suffers from a key limitation: many cells are suppressed to preserve con-

[1] This refers to the MSA File in the CBP dataset; however, the MSA identification codes correspond to CBSA codes.

[2] During our sample period, no casino hotel reports having employees; therefore, NAICS code 72110 does not appear in the NES data.

fidentiality (Eckert et al., 2021)[3]. To ensure sufficient CBSA-level coverage, we apply the following method to approximate employment figures[4]. First, for each CBSA in which a gambling-related NAICS code appears, if the employment value is nonzero, we retain it as-is. Second, for cases where the employment value is reported as zero, we estimate it by multiplying the number of establishments in each employment-size interval by the median value of that interval. For instance, in CBSA 11460, there are two establishments, one in the 1–4 interval and one in the 5–9 interval, so the estimated employment is calculated as $1 \times 2.5 + 1 \times 7 = 9.5$. For establishments falling within the highest interval (i.e., open-ended, 5,000 or more), we conservatively assign the lower bound value of 5,000. Third, we compare the estimated employment figure with the lower bound of the corresponding employment level flag and retain the value that is greater[5].

The NES dataset also contains limitations: numerous cells are suppressed to preserve confidentiality or to avoid quality issues. In such cases, the number of establishments is recorded as zero; however, the establishment flags indicate that the zero does not reflect true nonexistence. Instead, these suppressed values result from either unwillingness to disclose individual business information or concerns regarding data reliability. To address this issue, we implement the following imputation strategy. For observations with flag `D` (confidentiality concerns), we replace the zero with a value of 2[6]. For observations with flag `S` (reliability concerns), we substitute the zero with the state-level average for the corresponding industry category.

For geographic control variables, the U.S. Census Bureau does not provide CBSA-level data for the year 2000. To address this limitation, we aggregate county-level data up to the CBSA level. For ratio variables, we first aggregate the numerator and denominator separately across constituent counties, and then compute the ratio based on the aggregated values. For unemployment rates obtained from the U.S. Bureau of Labor Statistics, we apply the same aggregation procedure to the annual average data and implement a one-year lag in our regressions. For HPA, we use the CBSA-level HPI data directly from the FHFA.

[3]Another limitation discussed in Eckert et al. (2021) is the periodic reclassification of industries. However, since changes to gambling-related NAICS codes are infrequent and our sample period is short, this concern is minimal for our analysis.

[4]Eckert et al. (2021) proposes a solution for county-level data, but not for CBSA-level data. We adapt their core approach in a simplified form due to constraints that are difficult to satisfy at the CBSA level. Our primary strategy involves using the median value within employment intervals and verifying that the estimated employment remains within the bounds of the corresponding employment level flag interval.

[5]Refer to Tables 1 and 3 in Eckert et al. (2021) for definitions of employment intervals and employment level flag intervals, respectively. In our sample, all estimated values fall below the corresponding upper bounds of the employment level flags.

[6]We use 2 since the smallest observable nonzero value is 3. In untabulated results, substituting with 1 does not materially affect our findings.

Table A1 presents the summary statistics for all constructed gambling preference measures.

A.2 Validation Tests

To validate our alternative measures and explore their relation with CPRATIO, we do the following tests.

First, to examine the relationship between CPRATIO and both the alternative gambling preference measure and the two additional control variables, we conduct a univariate sorting test. CBSAs are sorted into quintiles according to their CPRATIO values, and we calculate equal-weighted quintile averages for the alternative measure of gambling preference as well as the control variables. Table A2 summarizes the results. For the alternative measure, we find that gambling industry employment tends to increase with CPRATIO, and the difference between the highest and lowest quintiles is statistically significant. This suggests that employment in the gambling industry captures a construct that is consistent with CPRATIO[7].

For the control variables, employment in casino hotels do not show a substantial increase with *CPRATIO*. This outcome may stem from two factors. First, the presence of casino hotels is influenced not only by local gambling demand but also by tourist-oriented gambling activity. As a result, local gambling preference does not necessarily correlate with casino hotel employment or establishment density. Second, exceptions in gambling legality, such as tribal casinos, permit casino hotel operations in certain regions even when local gambling preference is low. In contrast, the number of nonemployer gambling establishments tends to increase with *CPRATIO*. This pattern suggests that such establishments complement minor local gambling demand and provide an additional proxy for community-level gambling activity.

Second, to further examine whether *CPRATIO* can explain variation in the alternative gambling preference measure and the additional control variables, we estimate a series of OLS regressions. Each dependent variable is regressed on *CPRATIO* and CBSA-level local control variables with state and year fixed effects. All continuous variables are winsorized at the 0.5 percent level and standardized prior to analysis. Table A3 presents the regression results. The gambling industry employment is positively and significantly associated with *CPRATIO*, indicating that higher *CPRATIO* values correspond to greater levels of the supply-side

[7]Note that the unconditional correlation between the CPRATIO and gambling industry employment is not large. The reason is that although Catholics believe gambling is not in itself contrary to justice, Catholic bishops' conferences have opposed the expansion of gambling industry due to its social harms. Therefore, in areas with high proportion of Catholic residents, even CPRATIO is high, the gambling industry may not high as well.

indicator. This supports the use of *CPRATIO* as a proxy for local gambling preference. In contrast, the per capita employment in casino hotels is negatively correlated with *CPRATIO*, reinforcing our decision to treat casino hotel variables as controls for tourist-oriented gambling demand rather than proxies for local gambling preference. Finally, for nonemployer gambling establishments, which serve as a complementary measure for minor local gambling activity, we also observe a significantly positive coefficient on *CPRATIO*. This suggests that higher *CPRATIO* values are associated with increased presence of nonemployer gambling businesses, further corroborating *CPRATIO*'s relevance as an indicator of local gambling preference.

A.3 Additional Tests

In the main text, we reports the results that include per capita employment in the gambling industry only, we now include controls for employment in casino hotels to account for tourist-driven gambling demand and nonemployer gambling establishments to capture complementary gambling demand that may be excluded by the primary measures due to database limitations.

Table A4 presents the results of the regression. Columns (1) report specifications using per capita employment in the gambling industry only, as in the main text. The regression results indicate that employment metrics for the gambling industry are significantly positive, suggesting a strong link between local gambling preference and the likelihood of second-lien misrepresentation. By contrast, casino hotel variables, which reflect a mix of local and tourist demand, yield coefficients that are generally insignificant. Although nonemployer gambling businesses primarily reflect local demand, they tend to offer limited gambling access, and their associated coefficients are likewise insignificant. Overall, we find that per capita employment measure in the gambling industry, which is used as alternative proxy for local gambling preference, supports the hypothesis that second-lien misrepresentation is more prevalent in regions with elevated gambling preference.

A.4 Additional Illustration

In this section, we address several concerns related to the use of alternative measure of local gambling preference.

First, several limitations prevent us from using the alternative measure as the basis for our primary analysis. Although employment and establishment data in the gambling industry may more directly reflect local gambling preferences, these indicators represent supply-side dynamics rather than demand-side behavior. Furthermore, they are influenced by factors such as local legality and economic conditions, which introduces additional noise into the measurement. Another key limitation is the restricted coverage. The alternative measure span only 460 CBSAs, encompassing approximately 1,187 counties, and cover around 1 million loans. In contrast, CPRATIO includes data for nearly 3,000 counties and covers almost the entire sample of 3 million loans. This broader coverage is especially critical when analyzing lender-specific effects and borrower preference heterogeneity.

Second, we rely on CBSA-level data rather than county-level data to construct our gambling preference measures. The rationale for using a broader geographic and economic unit is that residents in most U.S. metropolitan areas can easily commute within the city. Consequently, individuals residing in counties with few gambling establishments may work or gamble in adjacent counties with greater access. As a result, low employment or establishment counts in a given county do not necessarily reflect low gambling preference among its residents. By contrast, CBSAs are sufficiently large that cross-CBSA commuting for work or gambling is relatively rare. Instances of inter-CBSA gambling, such as through tourism, are mostly captured by the casino hotel measures. Therefore, the CBSA level offers a more robust and stable proxy for local gambling preferences than county-level data.

Third, we restrict our sample to CBSAs where establishments exist in all three categories: gambling industry, casino hotels, and nonemployer gambling businesses. This selection improves the clarity and reliability of our gambling preference measures. In such areas, false zeros, resulting from legal prohibitions, data suppression, or missing supply, are minimized. In untabulated regressions, when we replace missing values with zeros and include all CBSAs, the coefficient on the gambling preference measure remains positive but becomes only marginally significant. Furthermore, when we split the sample into high and low per capita establishment groups, the coefficient is strongly significant in the high-establishment group but insignificant in the low group. This pattern suggests that second-lien misrepresentation does rise with elevated local gambling preference, whereas low values in the measure may reflect extraneous noise rather than genuinely low gambling tendencies.

Table A1
Descriptive Statistics for Alternative Gambling Preference Measures

	N	Mean	SD	P10	P25	P50	P75	P90
Gambling Industries (NAICS 7132, CBSA-year level)								
Employment (per 10,000 residents)	1,201	19.60	42.74	0.22	0.69	2.53	16.41	59.50
Casino Hotel (NAICS 72110, CBSA-year level)								
Employment (per 10,000 residents)	258	118.49	264.87	0.07	0.50	16.90	77.41	389.73
Nonemployer Gambling (NAICS 7132, CBSA-year level)								
Establishment (per 10,000 residents)	1,239	0.38	0.50	0.00	0.14	0.27	0.45	0.76

The table presents descriptive statistics for the alternative local gambling preference measure and its related controls, covering the sample period from 2005 to 2007. We report the number of observations, mean, standard deviation, 25th percentile, 50th percentile, and 75th percentile.

Table A2
Univariate Sorting

Quintile	Gambling Industries Employment	Casino Hotel Employment	Nonemployer Gambling Establishment
Low	16.2	114.20	0.28
Q2	11.77	140.31	0.35
Q3	18.67	49.83	0.43
Q4	26.47	95.67	0.40
High	24.92	219.91	0.46
High-Low	8.66	105.72	0.18
t-Statistic	2.02	1.72	4.01

The table presents univariate sorting results for three gambling preference related measures: per capita employment in the gambling industry (Gambling Industries Employment), per capita employment in the casino hotel industry (Casino Hotel Employment), and per capita number of nonemployer gambling establishments (Nonemployer Gambling Establishment). CBSAs are sorted into quintiles based on CPRATIO, and for each quintile, the equal-weighted average of the five gambling preference related measures is computed. The table reports White robust t-statistics for the difference in mean values between the High and Low CPRATIO quintiles.

Table A3
Alternative Measures and CPRATIO

	(1) Gambling Industry Employment	(2) Casino Hotel Employment	(3) Nonemployer Gambling Establishment
CPRATIO	0.103**	-0.161*	0.126***
	(0.043)	(0.084)	(0.038)
REL	-0.161***	-0.086	-0.159**
	(0.059)	(0.151)	(0.071)
Crime rate	-0.113***	-0.134	-0.028
	(0.040)	(0.131)	(0.038)
Republic	-0.005	-0.088	-0.212***
	(0.062)	(0.144)	(0.075)
Education	0.004	-0.819***	-0.094
	(0.055)	(0.241)	(0.070)
Income	-0.227***	1.056***	0.130
	(0.067)	(0.294)	(0.082)
Unemployment	0.018	-0.121	0.001
	(0.060)	(0.088)	(0.067)
HPA	0.017	0.124	0.085
	(0.044)	(0.092)	(0.052)
Urban	0.084	-0.452***	-0.030
	(0.067)	(0.119)	(0.052)
Total population	-0.072	-0.011	-0.017
	(0.048)	(0.126)	(0.061)
Married	0.067	-0.540***	0.004
	(0.045)	(0.167)	(0.059)
Over65	0.084**	0.045	0.034
	(0.040)	(0.114)	(0.051)
Male-female ratio	0.062	-0.000	-0.039
	(0.044)	(0.085)	(0.036)
Minority	0.040	0.192*	-0.042
	(0.045)	(0.116)	(0.048)
State FE	Y	Y	Y
Half-year FE	Y	Y	Y
Observations	1,015	227	1,042
Adj. R^2	0.330	0.541	0.281

The table shows OLS estimates from regressions where the dependent variables are the alternative gambling preference measure and related control variables. CPRATIO and other geographic controls are aggregated to CBSA level. All continuous variables are winsorized at the 0.5 percent level and then standardized. White robust standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A4
Alternative Gambling Preference Measures and Second-lien Misrepresentation

	(1)
Gambling Industry Employment	0.0030***
	(0.001)
Casino Hotel Employment	-0.0013
	(0.002)
Nonemployer Gambling Establishment	0.0010
	(0.002)
REL	0.0022
	(0.002)
Crime rate	0.0040**
	(0.002)
Republic	-0.0012
	(0.002)
Education	-0.0006
	(0.002)
Income	0.0006
	(0.002)
Unemployment	-0.0011
	(0.001)
HPA	0.0016
	(0.002)
Urban	0.0006
	(0.002)
Total population	0.0005
	(0.002)
Married	0.0001
	(0.001)
Over65	-0.0002
	(0.001)
Male-female ratio	0.0020
	(0.002)
Minority	-0.0025*
	(0.002)
Loan chars controls	Y
Originator FE	Y
State FE	Y
Half-year FE	Y
Observations	1,032,077
Adj. R^2	0.318

The table shows OLS estimates from regressions where the dependent variable takes a value of one if the loan is recognized as second-lien misreported and zero otherwise. The local controls are aggregated to CBSA level. All continuous variables are winsorized at the 0.5 percent level and then standardized. Standard errors clustered at the county level are reported in the parentheses. $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

B Diff-in-disc concerns

To validate our application of the diff-in-disc approach in comparing the number of loans and default rates between high and low gambling preference areas, we address the following concerns in this section.

B.1 Assumptions for Applying Diff-in-disc

According to Grembi et al. (2016), the validity of the diff-in-disc approach depends on three assumptions. First, all potential outcomes must be continuous around the cutoff. In our setting, this means no manipulation of FICO scores around the cutoff point of 620. Since the data on the distribution of FICO scores in the U.S. population is not available to us, we argue this assumption by referencing the findings in Keys et al. (2010). They found that the distribution of FICO scores across the population is smooth using data from an anonymous credit bureau. Additionally, by exploring the reversal of anti-predatory lending laws in Georgia and New Jersey, they found that borrowers were either unaware of the differential screening around the threshold or unable to quickly manipulate their FICO scores[8].

Second, in the absence of treatment, the effect of the confounding policy around the cutoff is constant over the "diff" part. To fulfill this assumption, we check whether the pattern holds for full documentation loans around FICO 620, which have no treatment but similar confounding factors. Table B1 shows the results for the number of loans (panel A) and default rate (panel B) estimated using full-documentation loans. For full-documentation loans, the number of loans shows a minimal jump, and the difference between high and low gambling preference areas is slight. This evidence indicates that without the ease of securitization, having a FICO score below or above 620 would not cause the number of loans to increase differently between high and low gambling preference areas. Additionally, there is no jump in the default rate for all types of loans, and the difference in the increase of the default rate between high and low gambling preference areas is minor. These findings illustrate that without the ease of securitization, lenders do not have lax screening standards for any type of loan and do not screen differently between high and low gambling preference areas.

Third, the effect of the treatment around the cutoff does not depend on the confounding policy. This assumption states that there should be no interaction between the treatment and the confounding policy. In other words, the pre-determined outcomes and covariates should have similar jumps (or no jumps) between high and low gambling preference areas. We test this assumption by estimating equation 6 with pre-determined outcomes and co-

[8] According to the rating agency (Fair Isaac), strategic manipulation of FICO scores is difficult.

variates as the dependent variable and also add other control variables in the regression. We choose the same bandwidth (10) and kernel (uniform) as the main tests. The results are reported in Table B2. Column (1) presents the results of loan characteristics, showing that most characteristics have similar jumps between high and low gambling preference areas. The proportions of negative amortization and second-home loans are significant, but we argue that this does not affect our conclusion for several reasons. First, although the absolute jump magnitude is different, the ratio of 620^+ to 620^- is close, i.e., negative amortization jumps from 6.74/2.97 percent to 15.82/6.61 percent (2.35/2.23 times) in high/low gambling preference areas. Second, the proportion of these specific loans is too low to have a meaningful effect on the number of loans with misrepresentation and the default rate (i.e., in this local range, second homes make up only 1.2 percent of all loans and only 0.6 percent of misreported loans). Column (2) reports the results of local characteristics, showing that most characteristics have similar jumps between high and low gambling preference areas. Although the proportions of highly educated residents, the proportions of older residents, and the male–female ratio are statistically significant, their quantitative importance is minor because the magnitudes are very small. In the local range, the average share of highly educated residents is 25.01 percent, the average share of older residents is 12.2 percent, and the average male–female ratio is 28.89 percent. The differences between high and low gambling preference areas are only –0.331 percent, 0.322 percent, and 0.003 percent, respectively.

B.2 Sensitivity to Regression Choices

Beyond these three assumptions, we also check whether the findings are sensitive to the inclusion of control variables, the choice of bandwidth, and the choice of estimation kernel. We estimate models with and without covariates, choosing bandwidths of 8, 10, and 12, and using uniform or triangular kernels. Table B3 reports the ratio of estimated values at FICO 620^+ to 620^- for the number of loans[9]. The magnitude of jumps is similar to the baseline results: the number of different types of loans increases significantly due to the ease of securitization, and the jump for loans with misreported seconds is smaller in high gambling preference areas than in low gambling preference areas. These results confirm the robustness of the findings that lenders do not play a larger role in increasing second-lien misrepresentation in high gambling preference areas. Table B4 shows that the default rate of loans with misreported seconds is small and insignificant in all cases. Moreover, although not significant, the jump in high gambling preference areas is smaller than the jump in low gambling preference areas in most cases. These robustness tests show that lenders in high

[9]Since the number of loans is counted at the level of gambling preference areas while control variables are at the level of counties, control variables could not be included in the tests.

gambling preference areas are not more inclined to increase such misrepresentation than lenders in low gambling preference areas.

B.3 Placebo Test

Lastly, we conduct a placebo test to evaluate the possibility that our results are driven by chance. Specifically, following Goodman and Mayer (2018), we implement the diff-in-disc estimations at false FICO score thresholds below and above 620. Our inferences on the lender's role come from two findings from the diff-in-disc estimations: (1) the number of loans with misreported seconds increases much less in high gambling preference areas at FICO 620, and (2) the default rate of loans with misreported seconds either shows a small and insignificant jump in high gambling preference areas or a jump that is smaller in high gambling preference areas than in low gambling preference areas. Thus, at these false thresholds, we expect to find (1) no systematic difference in the increase of the number of loans with misreported seconds between high and low gambling preference areas as in our baseline results, and (2) no situation where the default rate of loans with misreported seconds shows a meaningful upward jump in high gambling preference areas and a jump that is larger in high gambling preference areas than in low gambling preference areas. To maintain a similar setting and avoid potential jumps in the misrepresentation rate found in the literature, we conduct the tests within the range of 601 to 639[10]. For the number of loans, we exclude the range of 615 to 625 to stay sufficiently away from the true threshold that could cause noise in jump estimation. Figure B1 plots the cumulative density function of the 28 placebo point estimates for the number of loans with misreported seconds. All of the placebo coefficients are greater than our estimated coefficient (-2.063) and smaller than the absolute value of our estimated coefficient with a small magnitude. We also perform a t-test on the placebo coefficients to check the null hypothesis that the true mean equals 0, and the p-value of this test is 0.819, which fails to reject the null hypothesis. These placebo tests support the idea that the difference in discontinuities for the number of loans with misreported seconds between high and low gambling preference areas is not due to chance. For the default rate, we only exclude the true threshold since no jump exists at the true threshold, so estimations at the scores near it would not be affected. Figure B2 plots the cumulative density function of the placebo point estimates for the default rate of loans with misreported seconds. Most placebo coefficients are smaller than the absolute value of our estimated coefficient (0.052). The larger ones are also insignificant, and they show an insignificant small jump in high gambling preference areas. The t-test on the placebo coefficients that checks whether the

[10] From Figure 3 in Griffin and Maturana (2016) about the misrepresentation rate, we can see clear jumps at FICO 600 and 640.

true mean equals 0 provides a p-value of 0.638. These placebo tests support the idea that lenders did not have laxer screening in high gambling preference areas.

Table B1
Discontinuities of Full-documentation Loans Around the Credit Threshold

	(1) All	(2) High	(3) Low	(4) High-Low
Panel A. Number of loans (Est. 620+/620-)				
All loan	1.186	1.244	1.154	0.090
				(2.27)
Reported second	1.161	1.183	1.152	0.031
				(0.58)
Misreported second	1.251	1.325	1.216	0.109
				(1.03)
Panel B. Default rate				
All loan	0.007	0.001	0.009	-0.007
	(0.36)	(0.24)	(0.08)	(-0.58)
Reported second	-0.007	-0.026	0.002	-0.027
	(-0.39)	(-1.03)	(0.06)	(-0.83)
Misreported second	0.015	-0.016	0.029	-0.045
	(-0.49)	(-0.89)	(0.03)	(-1.12)

The table presents the results of the estimations in Table 7 and Table 8 using full-documentation loans. Panel A reports the estimates for the number of loans at each FICO score. Panel B reports the estimates for loans that become 90 days or more delinquent using the MBA method in the first three years after origination. Using local linear regressions of the RDD approach, we estimate the number of loans and the default rate (with control variables) at FICO 620^- and FICO 620^+ and compute the ratio of the estimated number of loans at FICO 620^+ to 620^-. Using the diff-in-disc approach, we estimate the difference between high and low gambling preference areas. We perform the estimation for all loans, loans with correctly reported simultaneous seconds, and loans with misreported simultaneous seconds. The results for loans in all areas, loans in high gambling preference areas, loans in low gambling preference areas, and the difference between high and low gambling preference areas are reported in columns (1) to (4), respectively. For discontinuities, bias-corrected t-values (standard errors clustered at the county level) following Calonico et al. (2014) are reported in parentheses. For the difference in the ratio of the number of loans/default rate between high and low gambling preference areas, t-statistics based on heteroskedasticity-consistent standard errors/standard errors clustered at the county level are reported in parentheses.

Table B2
Estimates of the discontinuities in observable covariates

	(1) loan characteristics		(2) local characteristics
Interest rate (%)	-0.006	REL	-0.005
	(0.041)		(0.003)
Balance	0.015	Crime rate	-3.650
	(0.013)		(4.730)
LTV (%)	0.170	Republic	0.000
	(0.367)		(0.014)
ARM	-0.010	Education (%)	-0.331*
	(0.013)		(0.199)
Negative amortization	0.025***	Income	0.005
	(0.007)		(0.007)
Option ARM	0.001	Unemployment (%)	0.046
	(0.002)		(0.067)
Prepayment penalty	0.005	HPA	-0.005
	(0.015)		(0.004)
Cash-out	0.003	Urban (%)	0.529
	(0.013)		(0.336)
No-cash-out	0.002	Total population (ln)	0.030
	(0.007)		(0.027)
Investment	0.007	Married (%)	-0.076
	(0.007)		(0.095)
Second-home	0.007**	Over65 (%)	0.322***
	(0.003)		(0.106)
		Male-female ratio	0.003**
			(0.001)
		Minority (%)	-0.415
			(0.309)

The table reports the results of diff-in-disc regression for pre-determined outcomes and covariates using Equation 6 with other controls. The dependent variables are the pre-determined outcomes and covariates. Column (1) shows the tests for loan characteristics, and column (2) shows the tests for borrower characteristics. As in the main test, the bandwidth is set to 10 and the kernel is uniform. Standard errors clustered at the county level are reported in parentheses. *$p < 0.10$, **$p < 0.05$, and ***$p < 0.01$.

Table B3
Sensitivity of Results for Number of Loans

	Uniform				Triangular			
	High	Low	All	H-L	High	Low	All	H-L
Bandwidth = 8								
All loan	2.230	2.242	2.237	-0.013	2.250	2.204	2.225	0.046
				(-0.13)				(0.44)
Reported second	5.469	4.693	4.999	0.776	5.761	5.235	5.446	0.526
				(0.96)				(0.56)
Misreported second	2.709	5.095	3.957	-2.385	2.496	5.128	3.826	-2.632
				(-2.85)				(-3.21)
Bandwidth = 10								
All loan	2.240	2.289	2.266	-0.049	2.240	2.234	2.237	0.006
				(-0.52)				(0.06)
Reported second	5.498	4.581	4.939	0.917	5.616	4.971	5.228	0.645
				(1.23)				(0.74)
Misreported second	2.763	4.826	3.838	-2.063	2.609	5.133	3.900	-2.524
				(-2.83)				(-3.20)
Bandwidth = 12								
All loan	2.199	2.245	2.224	-0.047	2.237	2.262	2.250	-0.025
				(-0.50)				(-0.27)
Reported second	5.209	4.602	4.846	0.607	5.520	4.845	5.114	0.676
				(0.91)				(0.84)
Misreported second	2.656	4.388	3.590	-1.732	2.668	4.971	3.858	-2.302
				(-2.60)				(-3.20)
Bandwidth = 14								
All loan	2.197	2.199	2.198	-0.002	2.222	2.257	2.241	-0.036
				(-0.02)				(-0.39)
Reported second	5.278	4.666	4.915	0.612	5.424	4.788	5.043	0.636
				(0.95)				(0.84)
Misreported second	2.515	3.940	3.275	-1.426	2.645	4.755	3.747	-2.110
				(-1.99)				(-3.18)

The table reports the sensitivity of RDD and diff-in-disc regression for the number of loans to the choice of bandwidth and estimation kernel. For the difference between high and low gambling preference areas, t-statistics based on heteroskedasticity-consistent standard errors are reported in parentheses.

Table B4
Sensitivity of Results for Default Rate of Loans with Misreported Second

	Uniform				Triangular			
	High	Low	All	H-L	High	Low	All	H-L
Bandwidth = 8								
No control	0.005	0.019	-0.004	-0.014	-0.007	-0.016	-0.031	0.009
	(-0.24)	(-0.83)	(-1.18)	(-0.23)	(0.81)	(-1.43)	(-0.89)	(0.13)
With control	0.029	0.042	0.031	-0.013	-0.002	-0.017	-0.008	0.015
	(-0.48)	(-1.34)	(-1.31)	(-0.26)	(0.42)	(-1.90)	(-1.06)	(0.24)
Bandwidth = 10								
No control	0.001	0.052	0.014	-0.052	-0.004	0.011	-0.014	-0.015
	(-0.07)	(-0.86)	(-1.15)	(-0.98)	(0.32)	(-1.15)	(-1.07)	(-0.25)
With control	0.017	0.077	0.046	-0.061	0.012	0.021	0.015	-0.008
	(0.12)	(-1.09)	(-0.84)	(-1.31)	(0.01)	(-1.56)	(-1.14)	(-0.16)
Bandwidth = 12								
No control	-0.022	0.065	0.014	-0.087	-0.003	0.033	0.000	-0.036
	(0.30)	(-0.30)	(-0.50)	(-1.71)	(0.09)	(-1.00)	(-1.14)	(-0.65)
With control	0.003	0.081	0.045	-0.078	0.014	0.046	0.029	-0.033
	(0.29)	(-0.08)	(0.04)	(-1.79)	(0.07)	(-1.26)	(-0.91)	(-0.68)
Bandwidth = 14								
No control	0.007	0.071	0.033	-0.064	-0.007	0.046	0.007	-0.053
	(-0.47)	(0.11)	(-0.72)	(-1.39)	(0.06)	(-0.59)	(-0.88)	(-1.03)
With control	0.023	0.082	0.054	-0.059	0.012	0.058	0.036	-0.046
	(-0.26)	(0.65)	(0.19)	(-1.40)	(0.09)	(-0.57)	(-0.43)	(-1.04)

The table reports the sensitivity of RDD and diff-in-disc regression for the default rate of loans with misreported seconds to the inclusion of control variables, the choice of bandwidth, and the choice of estimation kernel. For discontinuities, bias-corrected t-values (standard errors clustered at the county level) following Calonico et al. (2014) are reported in parentheses. For the difference between high and low gambling preference areas, t-statistics based on heteroskedasticity-consistent standard errors are reported in parentheses.

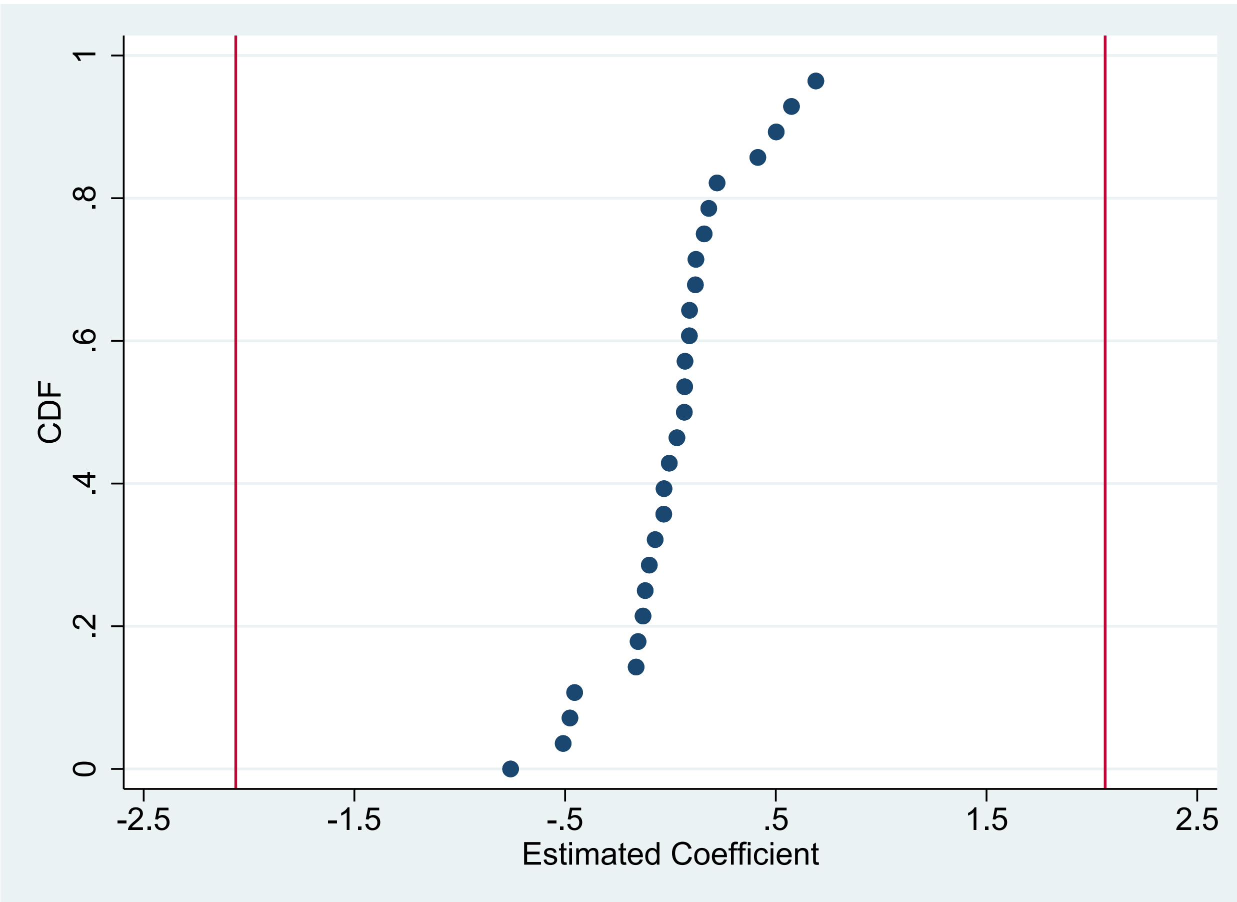

Figure B1
Placebo Tests for Number of Loans with Misreported Second
The figure plots the empirical c.d.f. of the estimated coefficient for the number of loans with misreported seconds from a set of diff-in-disc estimations at false FICO score thresholds below and above 620 (i.e., any score from 601 to 614 and any score from 626 to 639). The vertical lines show the benchmark estimate (-2.063) and its positive value (2.063).

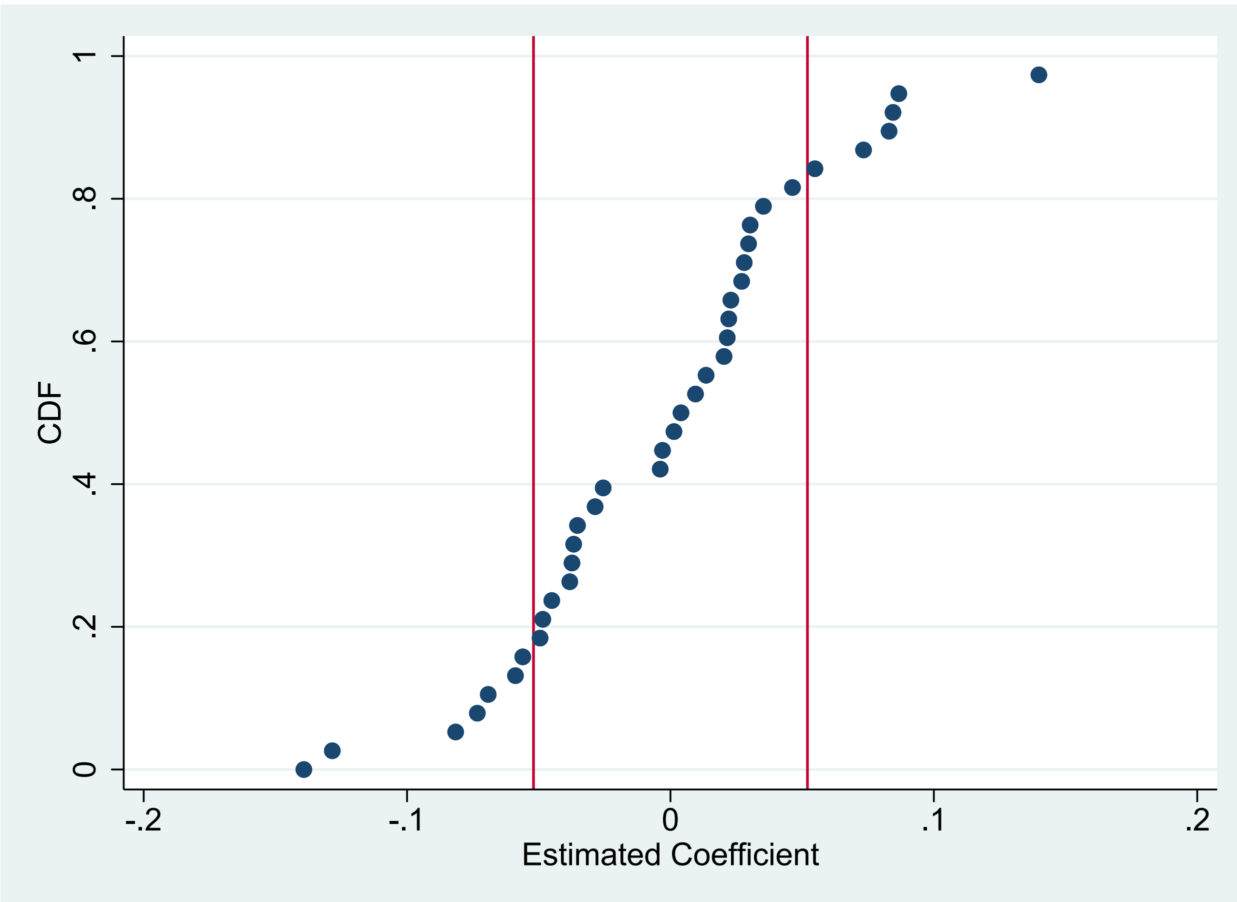

Figure B2
Placebo Tests for Default Rate of Loans with Misreported Second
The figure plots the empirical c.d.f. of the estimated coefficient for the default rate of loans with misreported seconds from a set of diff-in-disc estimations at false FICO score thresholds below and above 620. The vertical lines show the benchmark estimate (-0.052) and its positive value (0.052).

C Other Tables

In this section, we presents other tables mentioned in the main text.

Table C1
Local Structural Determinants Variables Correlation

		1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	CPRATIO	1.00													
2	REL	-0.01	1.00												
3	Crime rate	0.01	-0.13	1.00											
4	Republic	-0.28	0.07	-0.19	1.00										
5	Education	0.26	-0.09	0.14	-0.18	1.00									
6	Income	0.27	-0.11	-0.01	-0.05	0.64	1.00								
7	Unemployment	-0.06	-0.20	0.13	-0.17	-0.40	-0.34	1.00							
8	HPA	0.26	-0.29	0.03	-0.09	0.22	0.18	-0.21	1.00						
9	Urban	0.33	-0.01	0.49	-0.18	0.51	0.41	-0.17	0.15	1.00					
10	Total population	0.33	-0.18	0.47	-0.25	0.52	0.51	-0.08	0.23	0.74	1.00				
11	Married	-0.20	0.12	-0.44	0.43	-0.29	0.13	-0.14	-0.08	-0.40	-0.37	1.00			
12	Over65	-0.15	0.30	-0.26	0.07	-0.33	-0.42	0.03	0.00	-0.33	-0.37	0.23	1.00		
13	Male-female ratio	0.05	-0.21	-0.19	0.09	-0.01	0.11	-0.08	0.12	-0.17	-0.19	0.17	-0.21	1.00	
14	Minority	0.08	-0.09	0.43	-0.29	0.01	-0.15	0.22	0.13	0.20	0.23	-0.52	-0.29	-0.12	1.00

The table shows the correlation between local characteristics variables at the county-year level. Income and HPA at the county level are obtained from the same data sources as the ZIP code-level data, which are the U.S. Census Bureau and the Federal Housing Finance Agency, respectively. Unemployment is the average of the monthly unemployment rate within each year.